CS 700 Midterm Notes Mon 10/26, open book and notes, no electronics

1. Metrics (Lecture 2)

Good metrics are linear, reliable, repeatable, easy to use, consistent and independent. Reliable, repeatable, consistent and independent are required. Linear and easy to use are nice but not necessary.

MIPS $= \dfrac{n}{T_e \times 10^6}$, MFLOPS $= \dfrac{f}{T_e \times 10^6}$. MIPS fails reliable and consistent (CISC vs RISC, CPI). MFLOPS gives an integer program 0 and a FLOP isn't defined the same everywhere. SPEC normalizes to a basis machine and takes a geometric mean, which is the wrong mean for times.

Means metrics (clock rate, MIPS, MFLOPS) count what was done. Ends metrics (execution time) count what got accomplished. Execution time is the fundamental metric. Report wall clock and CPU time.

Speedup of system 2 over 1 is $S_{2,1} = \dfrac{T_1}{T_2} = \dfrac{R_2}{R_1}$. Relative change is $\Delta_{2,1} = \dfrac{T_1 - T_2}{T_2} = S_{2,1} - 1$. Precision $= \dfrac{\text{relevant retrieved}}{\text{retrieved}}$, recall $= \dfrac{\text{relevant retrieved}}{\text{all relevant}}$.

2. Summarizing data (Lecture 4)

Center

Mean $\bar{x} = \frac{1}{n}\sum x_i$, median is the middle sorted value (average the two middle ones when n is even), mode is the most frequent value or bin. Use the median when the data is skewed or has outliers, the mode for categorical data, and the mean when the total matters.

Harmonic $\bar{x}_H = \dfrac{n}{\sum 1/x_i}$ for rates whose numerator (work) is fixed.

Geometric $\bar{x}_G = \left(\prod x_i\right)^{1/n}$ for ratios that compound or normalized values. Arithmetic for times, sizes and anything whose sum means something.

The mean of a rate should track total work over total time. If every program does the same work F, then the right average rate is $\dfrac{nF}{\sum t_i}$, which is the harmonic mean of the rates.

Weighted forms, with $\sum w_i = 1$, $\bar{x}_A = \sum w_i x_i$, $\bar{x}_H = 1/\sum \frac{w_i}{x_i}$, $\bar{x}_G = \prod x_i^{w_i}$. Always $\bar{x}_H \le \bar{x}_G \le \bar{x}_A$, equal only when all values are equal.

Spread

Range $= x_{\max} - x_{\min}$. Sample variance $s^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1} = \dfrac{\sum x_i^2 - n\bar{x}^2}{n-1}$. Standard deviation s. Coefficient of variation COV $= s/\bar{x}$ (unitless). IQR $= Q_3 - Q_1$, SIQR $= \dfrac{Q_3 - Q_1}{2}$. Midhinge $= \dfrac{Q_1 + Q_3}{2}$. Use s with the mean, SIQR or IQR with the median.

Quantiles, course convention

Sort $x_{(1)} \le \cdots \le x_{(n)}$ and give $x_{(i)}$ the plotting position $f_i = \dfrac{i - 0.5}{n}$. For a quantile q, find i with $f_i \le q \le f_{i+1}$ and interpolate linearly between $x_{(i)}$ and $x_{(i+1)}$. Histogram bins by Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ on the slide (HW3 rounded instead, so say which you used). Excel's PERCENTILE on the slides interpolates at $1 + (n-1)p$, which gives slightly different quartiles.

Normal Q-Q plot

Plot $\left(z_i,\ x_{(i)}\right)$ with $z_i = \Phi^{-1}(f_i)$, or rescale to $m + s\,z_i$ so both axes share units. A straight line means normal. Ends bending away from the line mean heavy tails, and a curve bending one way means skew.

3. Confidence intervals (Lecture 6)

Mean

Large n (slides say $n > 30$) $\quad \bar{x} \mp z_{\alpha/2}\dfrac{s}{\sqrt{n}}$.

Small n, normal population $\quad \bar{x} \mp t_{[\alpha/2,\,n-1]}\dfrac{s}{\sqrt{n}}$.

All decks use upper-tail notation, $z_{\alpha/2}$ or $z[\alpha/2]$ is the value with area $\alpha/2$ above it, which the CI deck also calls the $(1-\alpha/2)$-quantile. Same number, e.g. 1.96 for 95%. The p.50 table labels them $z_{0.95}$, $z_{0.975}$. For $n = 15$ use df 14, $t = 2.145$.

Common levels $\quad 90\% \to 1.645$, $95\% \to 1.960$, $99\% \to 2.576$.

Batch means split a correlated series into batches and treat batch means as the sample. A 95% CI means 95% of intervals built this way cover μ. It does not mean μ has a 95% chance of being in this particular interval, and it says nothing about 95% of the data. The width shrinks like $1/\sqrt{n}$, so halving it takes four times as many runs.

How many measurements

For a half-width of e as a fraction of the mean, $n = \left(\dfrac{z_{\alpha/2}\,s}{e\,\bar{x}}\right)^2$, rounded up. Slide example $s = 2.14$, $\bar{x} = 7.94$, 90%, $e = 0.035$ gives $n = 160.5 \to 161$. The slide's "within 7%" is the full width, so the half-width is 3.5%.

Proportion

$p = m/n$, $\quad p \mp z_{\alpha/2}\sqrt{\dfrac{p(1-p)}{n}}$, valid when $np \ge 10$ (slide) and $n(1-p) \ge 10$ (study guide). Wald interval, poor for small n. Sample size with e as a RELATIVE error (slides) $n = \dfrac{z_{\alpha/2}^2\,p(1-p)}{(e\,p)^2}$. Slide example $e = 0.005$, $p = 0.1097$ gives 1.25 million 10 ms samples, 3.46 hours. With an absolute half-width e (measurement study guide) drop the p in the denominator. Slide example $m = 658$ of $n = 6000$, so $p = 0.1097$ and the 95% CI is 10.2% to 11.8%.

Variance (normal data only)

$\left(\dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},\ \dfrac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}\right)$. The larger χ^2 quantile goes in the left denominator.

4. Hypothesis testing (Lecture 9)

Steps state H_0 and H_1, choose α and one or two sides BEFORE the data, compute the statistic, compare with the critical value or find p, then report the effect with an interval.

$Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, $\quad t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $n-1$ df. Two-sided, reject if $|Z| > z[\alpha/2]$. One-sided, reject if $Z > z[\alpha]$. A two-sided test at α rejects exactly when the $(1-\alpha)$ CI excludes μ_0.

p is the probability of a statistic at least this extreme IF H_0 is true. It's not the probability that H_0 is true.

	H_0 true	H_0 false
Reject	Type I (α)	correct (power $1-\beta$)
Don't reject	correct	Type II (β)

Runs needed to detect a difference δ $\quad n = \left(\dfrac{(z[\alpha/2] + z[\beta])\,\sigma}{\delta}\right)^2$. At $\alpha = 0.05$ and power 0.80, $(1.96 + 0.84)^2 \approx 7.8$, so $n \approx 7.8\,\sigma^2/\delta^2$.

Not rejecting is not proof of no difference. With m tests at α each, expect about $m\alpha$ false positives, and Bonferroni uses α/m per test. Holm sorts the p values and compares the k-th smallest with $\alpha/(m-k+1)$, stopping at the first failure. Benjamini-Hochberg compares the k-th smallest with $k\alpha/m$ and controls the false discovery rate. Correlated samples count for less, e.g. 400 samples with $n_{\text{eff}} = 100$ halve Z. Statistical significance says nothing about whether the effect matters, so give the effect size and its CI.

5. Comparing two alternatives (Lecture 10)

Pick the test

1. Same unit (workload, benchmark) under both? Yes means paired, no means unpaired.
2. Paired with roughly normal differences, paired t on the differences.
3. Paired with outliers, Wilcoxon signed-rank if symmetric, sign test or bootstrap if skewed.
4. Unpaired and roughly normal, Welch's t.
5. Unpaired and skewed or ordinal, Mann-Whitney U or bootstrap.

For paired tests, normality is about the differences, not A or B.

Paired

$d_i = a_i - b_i, \quad \bar{d} \mp t[\alpha/2,\, n-1]\dfrac{s_d}{\sqrt{n}}, \quad n$ is the number of PAIRS.

Pairing removes the shared variation $2\rho\sigma_A\sigma_B$ from the variance of the difference, so it helps whenever $\rho > 0$.

Unpaired, Welch

$$s = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}, \quad (\bar{x}_A - \bar{x}_B) \mp t[\alpha/2,\, \nu]\, s,$$

$$\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}}$$

Use ν as is or round DOWN (more conservative). It lies between $\min(n_A, n_B) - 1$ and $n_A + n_B - 2$. Pooled t assumes equal variances and uses $n_A + n_B - 2$ df. Welch is the default.

Reading results

If the CI for the difference contains 0, you can't say they differ at that level. Overlapping single-system CIs do NOT mean no difference (HW6 Miller example), and checking overlap is not a test. Runs needed paired $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$, unpaired per group $n = 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$.

6. Measurement (Lecture 11)

A CI describes sampling error and nothing else. Repetition shrinks random error but never systematic error such as bias, a wrong clock or perturbation. Strategies event counting gives exact totals with tiny overhead but no context, tracing gives full detail and order but the most perturbation and data, sampling sets the overhead you choose but is statistical and misses rare events. Exactness, context and low overhead, you get two. Report the clock and its resolution. Finer observation means more perturbation. Timer rollover after $(2^n - 1)\, T_c$ for an n-bit counter with tick T_c. A single reading is off by up to one tick, so time k repetitions and divide, $(T_k - T_{\text{loop}})/k$. Overhead share of a measurement is about $2c/(T + 2c)$ for read cost c.

7. Research practice

Plagiarism per Luke's TR includes copied text without quotes even when cited, close paraphrase, reused figures, and self-plagiarism of your own published text. Cite ideas, quote words. Authorship goes to people who made an intellectual contribution.

8. More formulas from the handouts and study guides

Metrics

$T_e = \dfrac{n \times \text{CPI}}{f_{\text{clock}}}$ (not on the slides). GFLOPS and TFLOPS divide by 10^9 and 10^{12}. Jain's fairness $J = \dfrac{(\sum x_i)^2}{n \sum x_i^2}$ (HW2 only).

Summarizing

Quantile q sits at position $r = nq + 0.5$ in the sorted data, interpolate between $x_{(\lfloor r \rfloor)}$ and the next value. Tukey fences $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$, beyond them is an outlier. Work-weighted harmonic mean uses $w_i = F_i / \sum F_j$. Inverse CDFs for Q-Q plots, $u = f_i$ exponential with mean a, $-a \ln(1-u)$ (rate $\lambda = 1/a$) Pareto, $k(1-u)^{-1/a}$ geometric, $\lceil \ln(1-u)/\ln(1-p) \rceil$ (Summ p.64 writes $\ln u$, which is the random-variate form) normal, $z \approx 4.91\,[q^{0.14} - (1-q)^{0.14}]$, then $m + s\,z$. Feitelson puts the data on the x-axis, Setia and HW3 on the y-axis, so slope rules read reversed. $\ln 0.9 = -0.1054$, $\ln 0.7 = -0.3567$, $\ln 0.5 = -0.6931$, $\ln 0.3 = -1.2040$, $\ln 0.1 = -2.3026$.

Confidence intervals

Batch means with m batches $\bar{b} \mp t[\alpha/2,\, m-1]\, s_b/\sqrt{m}$. Sample size scales as $n_2 = n_1(h_1/h_2)^2$ for half-widths h. Proportion size with ABSOLUTE error e $n = z_{\alpha/2}^2\, p(1-p)/e^2$, use $p = 0.5$ with no pilot (95%, $e = 0.03$ gives 1068). Log transform build the CI on $\ln x$, then (e^L, e^U) is an interval for the median, centred on the geometric mean. Bootstrap percentile 95% CI is the 2.5th and 97.5th percentiles of the B resampled statistics. Monte Carlo $\bar{Y}(n) = (b-a)\frac{1}{n}\sum g(X_i)$, $\text{Var} = \text{Var}(Y)/n$.

Hypothesis testing

Two proportions $Z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$, pooled $\hat{p} = \dfrac{m_1 + m_2}{n_1 + n_2}$. McNemar (paired yes/no) $Z = \dfrac{b - c}{\sqrt{b + c}}$, only the disagreements b and c count. Paired $t = \bar{d}/(s_d/\sqrt{n})$. Power $\approx P\big[Z > z[\alpha/2] - |\delta|\sqrt{n}/\sigma\big]$. Family-wise error $1 - (1-\alpha)^m$, e.g. 23% at $m = 5$, 92% at $m = 50$. Simultaneous intervals use $z[\alpha/(2m)]$. Correlated samples $n_{\text{eff}} = n\dfrac{1-\rho}{1+\rho}$. Pocock cutoffs for peeking 2.18 (2 looks), 2.41 (5 looks). Chi-square goodness of fit $\sum (O-E)^2/E$ with $k - m$ df (m includes the 1 for the total plus fitted parameters).

Comparing alternatives

Pooled $s_p^2 = \dfrac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$, Cohen's $d = (\bar{x}_A - \bar{x}_B)/s_p$, paired $d_z = \bar{d}/s_d$. Signed-rank, large n $\mu_W = n(n+1)/4$, $\sigma_W^2 = n(n+1)(2n+1)/24$. Sign test $S \sim \text{Bin}(n, \frac{1}{2})$, $z = (S - n/2)/(\sqrt{n}/2)$. Mann-Whitney $U_1 = W_1 - n_1(n_1+1)/2$, $\mu_U = n_1 n_2/2$, $\sigma_U^2 = n_1 n_2(n_1 + n_2 + 1)/12$, $A_{12} = U_1/(n_1 n_2)$. All pairs of c systems $1 - (1-\alpha)^{c(c-1)/2}$ chance of a false positive.

Measurement

Wrapped counter ticks = end + 2^n − start. Instrumentation overhead $(T_{\text{inst}} - T_{\text{base}})/T_{\text{base}}$. Ratio error $(\sigma_R/R)^2 \approx (\sigma_A/A)^2 + (\sigma_B/B)^2$. Unpaired SE of a difference $\sqrt{SE_A^2 + SE_B^2}$. Nested runs, p processes with k iterations each $\text{Var}(\bar{x}) = \sigma_p^2/p + \sigma_i^2/(pk)$, cost $c_p p + c_i pk \le B$. Extra processes beat extra iterations.

Student t critical values, upper-tail area

$t[\alpha/2,\, \nu]$ for a two-sided $(1-\alpha)$ CI, e.g. 95% uses the .025 column, 90% uses .05.

df	.10	.05	.025	.01	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
80	1.292	1.664	1.990	2.374	2.639
100	1.290	1.660	1.984	2.364	2.626
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Square roots

2	1.414	27	5.196	52	7.211	77	8.775
3	1.732	28	5.292	53	7.280	78	8.832
4	2.000	29	5.385	54	7.348	79	8.888
5	2.236	30	5.477	55	7.416	80	8.944
6	2.449	31	5.568	56	7.483	81	9.000
7	2.646	32	5.657	57	7.550	82	9.055
8	2.828	33	5.745	58	7.616	83	9.110
9	3.000	34	5.831	59	7.681	84	9.165
10	3.162	35	5.916	60	7.746	85	9.220
11	3.317	36	6.000	61	7.810	86	9.274
12	3.464	37	6.083	62	7.874	87	9.327
13	3.606	38	6.164	63	7.937	88	9.381
14	3.742	39	6.245	64	8.000	89	9.434
15	3.873	40	6.325	65	8.062	90	9.487
16	4.000	41	6.403	66	8.124	91	9.539
17	4.123	42	6.481	67	8.185	92	9.592
18	4.243	43	6.557	68	8.246	93	9.644
19	4.359	44	6.633	69	8.307	94	9.695
20	4.472	45	6.708	70	8.367	95	9.747
21	4.583	46	6.782	71	8.426	96	9.798
22	4.690	47	6.856	72	8.485	97	9.849
23	4.796	48	6.928	73	8.544	98	9.899
24	4.899	49	7.000	74	8.602	99	9.950
25	5.000	50	7.071	75	8.660	100	10.000
26	5.099	51	7.141	76	8.718		

Standard normal $\Phi(z) = P(Z \le z)$

$P(Z > z) = 1-\Phi(z)$ and $\Phi(-z) = 1-\Phi(z)$. Critical values $z[.05] = 1.645$, $z[.025] = 1.960$, $z[.01] = 2.326$, $z[.005] = 2.576$, $z[.20] = 0.842$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

χ^2 quantiles, $P(\chi^2_\nu \le x)$

df	0.005	0.025	0.05	0.95	0.975	0.995
1	0.00	0.00	0.00	3.84	5.02	7.88
2	0.01	0.05	0.10	5.99	7.38	10.60
3	0.07	0.22	0.35	7.81	9.35	12.84
4	0.21	0.48	0.71	9.49	11.14	14.86
5	0.41	0.83	1.15	11.07	12.83	16.75
6	0.68	1.24	1.64	12.59	14.45	18.55
7	0.99	1.69	2.17	14.07	16.01	20.28
8	1.34	2.18	2.73	15.51	17.53	21.95
9	1.73	2.70	3.33	16.92	19.02	23.59
10	2.16	3.25	3.94	18.31	20.48	25.19
11	2.60	3.82	4.57	19.68	21.92	26.76
12	3.07	4.40	5.23	21.03	23.34	28.30
13	3.57	5.01	5.89	22.36	24.74	29.82
14	4.07	5.63	6.57	23.68	26.12	31.32
15	4.60	6.26	7.26	25.00	27.49	32.80
16	5.14	6.91	7.96	26.30	28.85	34.27
17	5.70	7.56	8.67	27.59	30.19	35.72
18	6.26	8.23	9.39	28.87	31.53	37.16
19	6.84	8.91	10.12	30.14	32.85	38.58
20	7.43	9.59	10.85	31.41	34.17	40.00
21	8.03	10.28	11.59	32.67	35.48	41.40
22	8.64	10.98	12.34	33.92	36.78	42.80
23	9.26	11.69	13.09	35.17	38.08	44.18
24	9.89	12.40	13.85	36.42	39.36	45.56
25	10.52	13.12	14.61	37.65	40.65	46.93
26	11.16	13.84	15.38	38.89	41.92	48.29
27	11.81	14.57	16.15	40.11	43.19	49.64
28	12.46	15.31	16.93	41.34	44.46	50.99
29	13.12	16.05	17.71	42.56	45.72	52.34
30	13.79	16.79	18.49	43.77	46.98	53.67
40	20.71	24.43	26.51	55.76	59.34	66.77
50	27.99	32.36	34.76	67.50	71.42	79.49
60	35.53	40.48	43.19	79.08	83.30	91.95

Every homework problem from HW0 and HW2 to HW6, in the form the exam is likely to copy. Each one has the question in short form, the method as numbered steps, the key numbers and his submitted answer, and the usual mistake. Every number below was recomputed in Python from the data files. Where the homework needed a computer there's a small version you can do with a pencil. Places where the submitted answer drifts from the slides or the source are marked **Check.**

Table values used in this section

$z[0.05]$	1.645	$t[0.05, 15]$	1.753
$z[0.025]$	1.960	$t[0.025, 15]$	2.131
$z[0.01]$	2.326	$t[0.005, 15]$	2.947
$z[0.005]$	2.576	$t[\cdot, 3]$ at .05/.025/.005	2.353/3.182/5.841
$\Phi(0.33)$	0.6293	$t[0.025, 8]$	2.306
$\Phi(0.5)$	0.6915	$t[0.05, 8]$	1.860
$\Phi(0.83)$	0.7967	$t[0.05, 6]$	1.943
$\Phi(0.9)$	0.8159	$t[0.05, 37]$	1.687
$\Phi(1.3)$	0.9032	$t[0.025, 2]$	4.303
$\Phi(1.8)$	0.9641	$\Phi(0.21)$	0.5832
$\Phi(2.0)$	0.9772	$\Phi(0.67)$	0.7486
$\Phi(2.5)$	0.9938	$\Phi(1.38)$	0.9162
$t[0.05, 5]$	2.015	$t[0.025, 4]$	2.776
$t[0.05, 24]$	1.711	$t[0.10, 37]$	1.304

$z[a]$ is the upper-tail value with area a above it, so $z[0.025] = 1.960$ (CI p.30). $\Phi(z)$ is the area below z. Handy roots are $\sqrt{50} = 7.071$, $\sqrt{12} = 3.464$, $\sqrt{3} = 1.732$.

Which homework to open for which exam question

If the exam asks you to	Model
compute a probability, mean or variance from a table	HW0 Q5 to Q9
judge a metric against the six criteria	HW2
pick and compute a mean (times, rates, ratios)	HW3 Q1
find quartiles, a histogram, a Q-Q plot	HW3 Q2
build a z or t CI and compare widths	HW4 P1
fix a CI for skewed data	HW4 P2
run a one-sample test, p-value, threshold	HW5 P1
set up H_0 and H_1 for a claim	HW5 P2
compare two systems, paired or Welch	HW6 P1, P2
audit a described experiment	HW4 to HW6 P3

Answer key at a glance (his submitted answers, all recomputed)

HW0 Q4	1/16, 15/16, 3/8
HW0 Q5	discrete, $\mu = 2.84$, Var 5.134, SD 2.27, COV 0.80
HW0 Q6	0.75, 0.425, 4/17 = 0.235, 17/23 = 0.739
HW0 Q7	mean 50, SD 23.09, 0.125, 0.375, 0.125
HW0 Q8	0.0873, 0.1841, 0.8159
HW0 Q9	0.321, 0.203, 0.309
HW3 Q1	GM 1.4294 (42.94%), AM 203.33 ms, HM 2485.66 tps
HW3 Q2	means 2.0142, 0.2083, s^2 1.0457, 0.0440, Q-Q r 0.9993, 0.904
HW4 P1	$\mu = 2.0142$, $\sigma^2 = 1.0447$, all six CIs contain μ, 95% widths 0.52 vs 1.21
HW4 P2	$\mu = 0.2083$, $\sigma^2 = 0.0440$, batched widest, bootstrap narrowest, 99% batched dips to -0.158
HW5 P1	$Z = 1.80$, fail to reject, $p = 0.0718$, threshold 2.843 h
HW5 P2	$H_1: \mu < 740$, $Z = -2.50$, reject, $p = 0.0062$
HW6 P1	paired, $\bar{d} = 0.611$, CI $(-10.77, 11.99)$, not shown
HW6 P2	Welch, $\nu = 37.89$, CI $(-1.540, 0.781)$, not shown

HW0. Probability refresher (ungraded, solved here)

HW0 Q1 and Q2. Independent events, and $P[A$ and $B]$

1. Independent means knowing B happened doesn't change the chance of A, so $P[A \mid B] = P[A]$.
2. Plug into the multiplication rule $P[A \cap B] = P[A \mid B]\, P[B]$.

Answer. $P[A \cap B] = P[A]\, P[B]$.

Usual mistake. Mixing up independent with mutually exclusive. Exclusive events with nonzero probability are never independent, since $P[A \cap B] = 0$. In the course, independence is what makes $\sigma/\sqrt{n}$ the right standard error (HT p.20, CI p.65).

HW0 Q3. $E[kX + mY]$ for constants k, m

1. Expectation is a weighted sum (Summ p.10), so it splits over sums and constants come out.

Answer. $k\,E[X] + m\,E[Y]$, with or without independence.

Usual mistake. Doing the same for variance. $\mathrm{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$ needs the correlation, and that term is exactly what pairing buys (Comp p.6). With $k = 1$, $m = -1$ you get $E[A - B] = E[A] - E[B]$, which is why $\bar{d} = \bar{x}_A - \bar{x}_B$ (Comp p.8).

HW0 Q4. Fair coin tossed 4 times. P(all heads), P(at least one tail), P(exactly 2H 2T)

1. There are $2^4 = 16$ equally likely sequences.
2. All heads is 1 sequence, $1/16 = 0.0625$.
3. At least one tail is the complement, $1 - 1/16 = 15/16 = 0.9375$.
4. Exactly 2 heads is $\binom{4}{2} = 6$ sequences, $6/16 = 3/8 = 0.375$.

Usual mistake. Adding probabilities instead of using the complement for "at least one". The same coin is the error model on CI p.12. Two $\pm E$ sources give $x - 2E, x, x + 2E$ with $1/4, 1/2, 1/4$. With 3 sources, $P[x + E] = \binom{3}{2}/8 = 3/8$, and many sources turn the binomial into a normal (CI p.15).

HW0 Q5. Requests per ms with $P_0, P_1, P_2, P_4, P_5, P_6, P_7$ = .18, .22, .14, .18, .12, .10, .06. Type? Why is $P_7 = 0.09$ invalid? Mean, variance, SD, COV

1. Whole-number values only, so it's discrete.
2. Listed values sum to 1.00, so $P_3 = 0$. With $P_7 = 0.09$ the sum is $1.03 > 1$, and probabilities must sum to exactly 1.
3. $E[X] = \sum xp = 0.22 + 0.28 + 0.72 + 0.60 + 0.60 + 0.42 = 2.84$.
4. $E[X^2] = 0.22 + 0.56 + 2.88 + 3.00 + 3.60 + 2.94 = 13.20$.
5. $\mathrm{Var} = 13.20 - 2.84^2 = 13.20 - 8.0656 = 5.1344$.
6. $\sigma = \sqrt{5.1344} = 2.266$. $\mathrm{COV} = \sigma/\mu = 2.266/2.84 = 0.798$ (Summ p.41).

Answer. Discrete, mean 2.84, variance 5.134, SD 2.27, COV 0.80.

Usual mistake. Dividing by $n - 1$ here. These are known probabilities of a population, so there's no $n - 1$. Also forgetting the missing P_3 and putting P_4's value at $x = 3$.

Small version. $X \in \{0, 1, 2\}$ with .25, .5, .25 gives $E = 1$, $E[X^2] = 1.5$, Var 0.5, SD 0.707, COV 0.707. COV is meaningless if the mean is 0 or negative (Summ p.41).

HW0 Q6. Cohort with Spring 10 (6 BS, 4 MS) and Fall 30 (17 BS, 13 MS). P(Fall), P(MS), P(Spring|MS), P(Fall|BS)

1. Total 40. MS $= 4 + 13 = 17$, BS-only $= 6 + 17 = 23$.
2. $P[\text{Fall}] = 30/40 = 0.75$, $P[\text{MS}] = 17/40 = 0.425$.
3. Conditioning shrinks the group. $P[\text{Spring} \mid \text{MS}] = 4/17 = 0.235$.
4. $P[\text{Fall} \mid \text{BS}] = 17/23 = 0.739$.

Usual mistake. Dividing by the wrong total, for example 4/10, which is $P[\text{MS} \mid \text{Spring}]$. The same direction mix-up is the p-value trap, since $P[\text{data} \mid H_0] \neq P[H_0 \mid \text{data}]$ (HT p.13 to 14).

HW0 Q7. $X \sim U(10, 90)$. Mean, SD, $P[X < 20]$, $P[X > 60]$, $P[20 < X < 30]$

1. Mean $(a + b)/2 = 50$. Variance $(b - a)^2/12 = 6400/12 = 533.3$, SD $= 80/\sqrt{12} = 80/3.464 = 23.09$.
2. Density is flat at 1/80, so each probability is a length over 80.
3. $10/80 = 0.125$, $30/80 = 0.375$, $10/80 = 0.125$.

Usual mistake. Using $(b-a)/12$ or forgetting the square. Rounding error of a timer is uniform on $\pm\Delta$, so its SD is $2\Delta/\sqrt{12}$ (quantization, CI p.9 to 10).

HW0 Q8. $Z \sim N(0, 1)$. $P[0.9 < Z < 1.3]$, $P[Z > 0.9]$, $P[Z > -0.9]$

1. The table gives $\Phi(1.3) = 0.9032$ and $\Phi(0.9) = 0.8159$.
2. $0.9032 - 0.8159 = 0.0873$. Upper tail $1 - 0.8159 = 0.1841$.
3. Symmetry gives $P[Z > -0.9] = P[Z < 0.9] = 0.8159$.

Usual mistake. Answering $P[Z > -0.9]$ with 0.1841. Sketch the curve and shade before subtracting.

HW0 Q9. Highs $\sim N(85, 6^2)$. $P[83 < X < 88]$, $P[X > 90]$, $P[X < 82]$ with z-scores

1. $z = (x - 85)/6$. 83 gives -0.33 and 88 gives 0.5, so $0.6915 - (1 - 0.6293) = 0.3208$.
2. 90 gives 0.83, so $1 - 0.7967 = 0.2033$.
3. 82 gives -0.5, so $1 - 0.6915 = 0.3085$.

Answer. About 0.321, 0.203, 0.309 (unrounded z gives 0.322 and 0.202).

Backwards version. The hottest 5% of days start at $85 + 1.645(6) = 94.9$.

HW2. Beyond performance metrics (report)

HW2. Pick metrics from your area, evaluate them on the lecture's criteria, propose extra criteria (Metrics p.33)

1. List the six criteria (Metrics p.7 to 12). Linear (2x metric means 2x performance). Reliable (A scores higher means A really is better). Repeatable (same experiment, same value). Easy to use (easy to measure correctly). Consistent (same units and definition on every system). Independent (vendors can't bend it).
2. Say which are required. Reliable, repeatable, consistent and independent are required, while linear and easy to use are nice to have (Metrics p.13).
3. Classify means versus ends. Means-based counts what was done, useful or not. Ends-based counts what got accomplished (Metrics p.27 to 28).
4. For each metric, give a verdict per criterion with one concrete failure.
5. Propose extra criteria your area needs.

His answer. Area is low-latency video streaming. Metrics were average throughput and the QoE score (Yin et al. 2015, chunk quality minus a switch penalty minus a stall penalty, as a weighted sum).

Criterion	Throughput	QoE score
Linear	Yes	No
Reliable	No	No
Repeatable	Partly	Partly
Easy to use	Yes	Yes
Consistent	Partly	No
Independent	No	No

- Throughput isn't reliable because a sender can inflate it by filling a deep queue (bufferbloat), the same failure as MIPS (Metrics p.17). It isn't independent because ISPs advertise it. Repeatability is partial because Pantheon found algorithm rankings change with the path. It's a means metric used for ends claims, since 8 Mbps with two stalls beats 5 Mbps with none.
- QoE isn't consistent because each paper picks its own weights (Yin's stall weight 3000, Pensieve's three variants), the MFLOPS problem. It isn't linear since it adds unlike units. It isn't reliable because the linear form was never validated against subjective tests (Peroni and Gorinsky 2024). It isn't independent because authors set the weights they're judged under.
- Extra criteria. Externality, meaning what a flow does to others on the link, which Jain's index $(\sum x_i)^2/(n\sum x_i^2)$ hides. Report the distribution and the tail, not just the mean. State the conditions the metric was checked over.

By hand. Jain's index for 90 and 10 Mbps is $100^2/(2(8100 + 100)) = 10000/16400 = 0.61$. Precision and recall from Metrics p.32. 30 retrieved, 20 relevant, 40 relevant missed gives precision $20/30 = 0.67$ and recall $20/60 = 0.33$.

Check. The repeatability paragraph says Puffer's 95% CI on stall ratio was "around 10 to 17 percent of the mean" after "1.75 years" per scheme. Those numbers are from the arXiv preprint. The NSDI 2020 version he cites (and quotes in HW4) says ±13% to ±21% with at least 2.5 years per scheme. The argument stands, but the two homeworks quote different versions.

Usual mistake. Saying a metric "fails" without naming the failure. On the exam, MIPS is repeatable, easy and independent but not linear, reliable or consistent (Metrics p.17).

How to write it on the exam.

1. Name the metric and what it counts, then say means or ends.
2. Go down the six criteria in order, one line each, with a yes, no or partly and a concrete reason.
3. Flag which failures matter most, the required four (Metrics p.13).
4. Add one criterion your area needs and say why the six miss it.

Exam variant. MIPS $= n/(T_e \times 10^6)$ (Metrics p.16). A CISC machine runs 50M instructions in 1 s (50 MIPS) and a RISC machine runs 100M simpler instructions for the same program in 0.8 s (125 MIPS). MIPS ranks RISC 2.5x better, but the real speedup is $1/0.8 = 1.25$x. So MIPS isn't linear, and with different instruction sets it isn't consistent either. Execution time is the ends-based alternative (Metrics p.27 to 28).

HW3. Choosing a mean, quartiles, histograms, Q-Q plots

HW3 Q1a. Five stacked features improve quality 10, 50, 70, 85, 15%. Mean improvement per feature, and which mean?

1. Each feature builds on the last, so effects multiply. Use the geometric mean (Summ p.30, p.32).

2. Ratios 1.10, 1.50, 1.70, 1.85, 1.15. Product $1.65 \to 2.805 \to 5.18925 \to 5.9676$.
3. Fifth root (Summ p.23), $5.9676^{1/5} = 1.4294$.

Answer. GM 1.4294, so 42.94% per feature. Matches his main.tex.

Usual mistake. Taking the GM of the percentages (10, 50, ...) instead of the ratios, or forgetting to subtract 1 at the end.

By hand. Gains of 100, 300, 700% are ratios 2, 4, 8. Product 64, $\sqrt[3]{64} = 4$, so 300% per version. The AM of the ratios, 4.67, would compound to about 102, not 64.

HW3 Q1b. Compiler times 150, 200, 120, 300, 270, 180 ms. Mean, and which?

1. These are times, so the mean should be proportional to total time. Arithmetic mean (Summ p.25, p.27).
2. $1220/6 = 203.33$ ms.

Usual mistake. Using the harmonic mean because it's "performance". Times take the AM. Weighted version (Summ p.33) with times 2, 5, 10 s and weights .5, .3, .2 gives $1 + 1.5 + 2 = 4.5$ s.

HW3 Q1c. DB at 2500, 3500, 3700, 1800, 2000 tps. Mean, and which?

1. Rates have time in the denominator, so the mean should be inversely proportional to total time. Harmonic (Summ p.25, p.28).
2. $\sum 1/x_i = 0.000400+0.000286+0.000270+0.000556+0.000500 = 0.0020115$.
3. $x_H = 5/0.0020115 = 2485.66$ tps (Summ p.22).

Usual mistake. The AM of rates (2700 here) overstates. The HM is right only if every run did the same work F (Summ p.28).

By hand. Rates 2, 3, 6 give $1/2 + 1/3 + 1/6 = 1$, so $x_H = 3$. The AM would say 3.67.

Exam variant, unequal work. Program 1 does 100 Mops at 50 Mops/s (2 s), program 2 does 300 Mops at 100 Mops/s (3 s). The plain HM gives $2/(1/50 + 1/100) = 66.7$, but the true rate is total work over total time, $400/5 = 80$ Mops/s. The plain HM assumes equal work per run (Summ p.26, p.28).

HW3 Q2a,b. For normal.txt ($n = 1000$) and not-normal.txt ($n = 870$), mean, variance, median, Q1, Q2, Q3

1. Mean $\frac{1}{n}\sum x_i$. Sample variance with $n-1$, one-pass form $s^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}$ (Summ p.39).
2. Sort. The i-th value sits at $f_i = (i - 0.5)/n$. Read $Q(0.25), Q(0.5), Q(0.75)$ with linear interpolation (Summ p.59).

	normal	not-normal
Mean	2.0142	0.2083
Variance $(n-1)$	1.0457	0.0440
Min	−1.6554	0.0007
Q_1	1.2680	0.0587
Median $= Q_2$	2.0322	0.1429
Q_3	2.6927	0.2865
Max	5.2128	1.2957

not-normal is right-skewed. The mean 0.208 sits above the median 0.143, and $Q_3 - Q_2 = 0.144$ against $Q_2 - Q_1 = 0.084$ (Summ p.48). For skewed data report SIQR $= (Q_3 - Q_1)/2 = 0.114$ (Summ p.45, p.49).

By hand. Data 7, 1, 11, 5, 9, 3. Sorted 1, 3, 5, 7, 9, 11, mean $36/6 = 6$. $\sum x^2 = 286$, $(\sum x)^2/n = 216$, $s^2 = 70/5 = 14$, $s = 3.742$. The f_i are .083, .25, .417, .583, .75, .917. So $Q_1 = x_{(2)} = 3$, $Q_3 = x_{(5)} = 9$, and the median halfway between 5 and 7 is 6. With $n = 10$, $Q_1 = x_{(3)}$ and $Q_3 = x_{(8)}$ exactly.

Exam variant with interpolation. Data 2, 4, 4, 5, 7, 9, 10, 12 ($n = 8$). The f_i are .0625, .1875, .3125, .4375, .5625, .6875, .8125, .9375. Q_1 at .25 sits between $x_{(2)} = 4$ and $x_{(3)} = 4$, so 4. The median sits halfway between 5 and 7, so 6. Q_3 at .75 sits halfway between $x_{(6)} = 9$ and $x_{(7)} = 10$, so 9.5. SIQR $= (9.5 - 4)/2 = 2.75$.

Usual mistake. Dividing by n for a sample variance, or taking Q_1 as $x_{(n/4)}$ without the $(i - 0.5)/n$ positions.

HW3 Q2c. Histogram of each

1. Equal-width bins over the range, plot relative frequency (Summ p.54).
2. Bins from Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ (Summ p.54). $\log_2 1000 = 9.97$ and $\log_2 870 = 9.76$, so $k = 10$ for both.

Check. His main.tex says 11 bins for both because analyze.py used round$(1 + \log_2 n)$, which rounds 10.97 and 10.76 up. The slide floors, giving 10. No conclusion changes, but use the floor on the exam.

By hand. For six points, $k = \lfloor 1 + 2.58 \rfloor = 3$, range 10, width 3.33, counts 2, 2, 2. Bin width changes the shape (Summ p.55 to 56), so judge shape with the Q-Q plot.

HW3 Q2d. Quantile plot

1. Plot $(f_i, x_{(i)})$ with $f_i = (i - 0.5)/n$ (Summ p.59).
2. normal.txt gives an S shape. not-normal.txt stays flat, then shoots up near $f = 1$, the long right tail.

By hand. In the Summ p.60 table ($n = 10$), $Q(0.80)$ is halfway between $x_{(8)} = 44.5$ and $x_{(9)} = 50.3$, so 47.4.

HW3 Q2e. Compare with a normal using a Q-Q plot

1. For each f_i find z_i with $\Phi(z_i) = f_i$, from the table or $z_i = 4.91[f_i^{0.14} - (1 - f_i)^{0.14}]$ (Summ p.69).
2. Rescale $x_i = m + sz_i$ with the sample mean and SD, so normal data falls on the 45° line. Unscaled, the intercept is the mean and the slope the SD (Summ p.70).
3. Plot theoretical against observed (Summ p.63) and read the shape (Summ p.71 to 73).

Answer. normal.txt follows the line with correlation 0.9993, so it's consistent with normal. not-normal.txt sits above the line at both ends and below in the middle, correlation 0.904. The normal predicts about −0.47 at the low end but the data never go below 0, and the top values are far too large. Right-skewed, not normal.

By hand. Data 1, 3, 5, 7, 9, 11 with $m = 6$, $s = 3.742$. z at .083, .25, .417 is $-1.38, -0.67, -0.21$. Theoretical values $6 - 3.742(1.38) = 0.84$, then 3.49, 5.21, 6.79, 8.51, 11.16. Pairs (0.84, 1), (3.49, 3) and so on are all within 0.5 of the line, so normal. Skewed version 1, 1, 1, 2, 3, 10 has $m = 3$, $s^2 = 62/5 = 12.4$, $s = 3.52$, theoretical $-1.86, 0.64, 2.26, 3.74, 5.36, 7.86$. Ends above the line and middle below is the right-skew pattern.

Usual mistake. Plotting f_i instead of z_i on the x-axis, or reading "longer tails" (Summ p.71) when the shape is asymmetric (Summ p.73).

HW4. Confidence intervals

HW4 1.1 and 2.1. The file is the population. Population mean and variance

1. Divide by N, not $N-1$. The $n-1$ is for estimating from a sample (Summ p.39).
2. Converting HW3's value, $1.0457 \times 999/1000 = 1.0447$.

Answer. normal has $\mu = 2.0142$, $\sigma^2 = 1.0447$. not-normal has $\mu = 0.2083$, $\sigma^2 = 0.0440$ (the same values as CI p.32).

HW4 1.2. Sample of 50 from normal.txt. 90, 95, 99% CIs. Do they contain μ?

1. $n > 30$, so $\bar{x} \pm z[\alpha/2]\, s/\sqrt{n}$ (CI p.27, p.31).
2. $\bar{x} = 2.2093$, $s = 0.9443$, SE $= 0.9443/7.071 = 0.1335$.
3. Half-widths $1.645(0.1335) = 0.220$, $1.96(0.1335) = 0.262$, $2.576(0.1335) = 0.344$.

Answer. (1.9897, 2.4290), (1.9476, 2.4710), (1.8653, 2.5533). All contain 2.0142.

By hand. $n = 100$, $\bar{x} = 10$, $s = 2$ gives SE 0.2 and $10 \pm 0.392 = (9.61, 10.39)$. Runs for $\pm 5\%$ at 95% (CI p.44) are $n = (1.96 \cdot 2/(10 \cdot 0.05))^2 = 7.84^2 = 61.5$, so 62.

HW4 1.3. Sample of 16. Same three CIs

1. Small n and a normal population, so $\bar{x} \pm t[\alpha/2, n-1]\, s/\sqrt{n}$ with 15 df (CI p.37).
2. $\bar{x} = 2.2911$, $s = 1.1332$, SE $= 1.1332/4 = 0.2833$.
3. Half-widths $1.753(0.2833) = 0.497$, $2.131(0.2833) = 0.604$, $2.947(0.2833) = 0.835$.

Answer. (1.7944, 2.7877), (1.6872, 2.8949), (1.4562, 3.1259). All contain μ.

By hand. $n = 9$, $\bar{x} = 20$, $s = 6$ gives SE 2 and $20 \pm 2.306(2) = (15.39, 24.61)$. If σ is given, use z even for small n (CI p.27).

Exam variant from raw values. Data 8, 10, 12, 14. Mean 11, deviations $-3, -1, 1, 3$, $s^2 = 20/3 = 6.67$, $s = 2.582$, SE $= 2.582/2 = 1.291$. With $t[0.025, 3] = 3.182$ the 95% CI is $11 \pm 4.11 = (6.89, 15.11)$.

HW4 1.4. Compare widths from 1.2 and 1.3, explain with sample size and distribution

1. The 95% widths are 1.2077 ($n = 16$) and 0.5235 ($n = 50$), ratio 2.31.
2. Width is proportional to crit $\times\, s/\sqrt{n}$, so split the ratio into factors.
3. $\sqrt{50/16} = 1.77$ from sample size (CI p.39). $2.131/1.960 = 1.09$ from t having heavier tails (CI p.35). $1.1332/0.9443 = 1.20$ from luck in s.
4. Check, $1.77 \times 1.09 \times 1.20 = 2.31$.

Usual mistake. Blaming only n. The question asks for the critical value too. Halving a width takes 4 times the data.

HW4 2.2 and 2.3. Repeat with not-normal.txt

1. $n = 50$. The data aren't normal but the CLT makes $\bar{x}$ close to normal, so z is fine (CI p.29, p.67). $\bar{x} = 0.2168$, $s = 0.2278$, SE 0.0322.
2. $n = 16$. $\bar{x} = 0.2406$, $s = 0.2040$, SE $= 0.0510$, t with 15 df.

Level	$n = 50$ (z)	$n = 16$ (t_{15})
90%	(0.1638, 0.2698)	(0.1512, 0.3300)
95%	(0.1537, 0.2800)	(0.1319, 0.3493)
99%	(0.1339, 0.2998)	(0.0903, 0.3909)

All six contain 0.2083.

Usual mistake. Trusting the t interval at $n = 16$ on skewed data. The CI p.67 example, $n = 16$ from a skewed population, covers only about 91% at nominal 95% and runs too narrow.

HW4 2a. Batching. 16 values in draw order into 4 batches of 4, CI from the batch means with t on 3 df

1. Average groups of 4 or more to push toward normal, the CLT by hand (CI p.68). Keep draw order, since sorting would make each batch a different slice.
2. Batch means 0.4212, 0.2368, 0.0904, 0.2139. Mean $0.9623/4 = 0.2406$, same as raw.
3. $\sum x^2 = 0.2874$, $(\sum x)^2/4 = 0.2315$, $s^2 = 0.0559/3 = 0.0186$, $s = 0.1365$, SE $= 0.1365/2 = 0.0683$.
4. Half-widths 2.353, 3.182, 5.841 times 0.0683 give 0.161, 0.217, 0.399.

Answer. (0.0799, 0.4012), (0.0233, 0.4578), $(-0.1581, 0.6393)$. All contain μ.

By hand. 12 values into 3 batches with means 2, 4, 6. Mean 4, $s^2 = 8/2 = 4$, SE $2/\sqrt{3} = 1.155$, and the 95% CI is $4 \pm 4.303(1.155) = 4 \pm 4.97$. Batching needs lots of data.

HW4 2b. Bootstrap with $B = 10000$, seed 12345

1. Treat the sample as the population. Draw 16 with replacement, take the mean, repeat B times (CI p.69).
2. Sort the means. 95% runs from the 2.5th to the 97.5th percentile, 90% from the 5th to the 95th, 99% from the 0.5th to the 99.5th.

Answer. (0.1618, 0.3242), (0.1493, 0.3407), (0.1230, 0.3705).

By hand. Sample $\{2, 4, 9\}$. Resamples $\{2, 2, 9\}$ mean 4.33, $\{4, 9, 9\}$ mean 7.33, $\{2, 4, 4\}$ mean 3.33, $\{9, 2, 4\}$ mean 5. Repeat thousands of times, sort, read the percentiles.

Usual mistake. Resampling without replacement gives the same mean every time. If the 16 values missed the tail, every resample misses it too (CI p.69 to 70).

HW4 2c. Compare raw, batched and bootstrap at $n = 16$

	Raw t	Batched	Bootstrap
90% width	0.179	0.321	0.162
95% width	0.217	0.435	0.191
99% width	0.301	0.797	0.248

1. All nine contain 0.2083.
2. Batching is widest and bootstrap narrowest at every level, as the slides predict (CI p.68 says about 50% wider, CI p.70 calls bootstrap the narrowest). Here batching was about twice as wide because the batch means were spread out.
3. Batching widens because df fall from 15 to 3, so t goes from 2.131 to 3.182, and here the SE also rose from 0.0510 to 0.0683. The cost is precision and any statement about individual values.
4. The 99% batched interval reaches -0.158, impossible for positive data. Too few df at high confidence.
5. Coverage is a long-run rate over repeated samples (CI p.33). One hit is consistent with 95% and with 91%, so you'd have to repeat the draw and count.

Which fix when. Correlated observations, batch. Median or a multiplicative process, transform. Mean with independent data, bootstrap (CI p.70).

The third fix by hand, transform. HW4 skipped it, but it's the other response on CI p.68. Data 1, 10, 100. Take $\log_{10}$ to

get 0, 1, 2, with mean 1, $s = 1$, SE $= 1/\sqrt{3} = 0.577$. With $t[0.025, 2] = 4.303$ the log-scale CI is $1 \pm 2.48 = (-1.48, 3.48)$. Back-transforming gives $(0.033, 3048)$, centred on $10^1 = 10$. That's an interval for the median (here the geometric mean), not the mean, which is 37 (CI p.68 trap).

HW4 Problem 3. Confidence intervals in the wild, 8 questions

Checklist.

1. Paper and why.
2. Locate the first interval or error bar. Finding none is a finding.
3. Estimator. Mean, proportion, ratio, quantile, GM? Which lecture formula fits?
4. n. Justified, or just asserted?
5. Assumption. What does the interval assume? If skewed with small n, which of batch, transform, bootstrap (CI p.68)?
6. Independence. What's the unit, and are the units really independent (CI p.66)?
7. What the interval excludes. Systematic error, setup, population (CI p.17).
8. Interpretation. Do they claim "95% probability the true value is inside", which needs a credible interval?

His paper. Yan et al., Puffer, NSDI 2020.

1. Video streaming, already cited in HW2, and no Part I entry fits.
2. Sec. 3.4, "95% CI on stall ratio between ±13% and ±21% of the mean". Fig. 9 error bars are 95% CIs.
3. Stall ratio is a ratio, so bootstrap. Average SSIM is a duration-weighted mean with a z interval.
4. 637,189 streams from 54,612 IPs, about 2.5 to 2.8 stream-years per scheme. That's whatever 8 months gave, not planned. The "2 stream-years to separate a 15% difference" is unexplained.
5. Stalls are very skewed (4% of streams stalled). Bootstrap fits, but B and the interval type aren't stated.
6. Randomized by session but the bootstrap resamples streams, which share a connection. The intervals are likely too narrow. Resample sessions.
7. One Stanford server, self-selected viewers, no Safari. Says nothing about other users or servers.
8. No credible-interval claim. But they compare schemes by error-bar overlap, which isn't a test.

Exam scenario. "Mean RPC latency was 4.2 ms ± 0.1 ms (95% CI) from 10,000 requests in one run on one machine, so there is a 95% chance the true mean is between 4.1 and 4.3." Walk the checklist.

3. Estimator is a mean, z formula since n is large (CI p.31).
4. $n = 10{,}000$ is asserted, and it counts requests, not runs.
5. Latency is skewed, but for a mean at this n the CLT is fine (CI p.29).
6. The real problem. Consecutive requests share caches, queues and load, so they aren't independent and the interval is too narrow (CI p.66). Batch the requests or do several runs with one mean per run (CI p.70, HT p.22).
7. One machine, one run. Systematic error and the setup aren't in the interval (CI p.17).
8. "95% chance the true mean is inside" is a credible-interval claim. A CI only says the procedure covers μ in 95% of repeats (CI p.33).

HW5. Hypothesis testing

HW5 1.1. $\sigma = 0.8$ h known, $n = 36$, $\bar{x} = 2.74$. Test $H_0 : \mu = 2.5$ at $\alpha = 0.05$

1. $H_0 : \mu = 2.5$, $H_1 : \mu \neq 2.5$. No direction is named, so two-sided (HT p.5).
2. σ is known, so $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ (HT p.8). SE $= 0.8/6 = 0.1333$.
3. $Z = 0.24/0.1333 = 1.80$.
4. Reject if $|Z| > 1.960$. Since $1.80 < 1.96$, fail to reject.
5. Interval check (HT p.9). $2.74 \pm 1.96(0.1333) = (2.479, 3.001)$ contains 2.5.

Answer. $Z = 1.80$, fail to reject. Matches main.tex.

Small version. $n = 16$, $\sigma = 0.8$, $\bar{x} = 2.9$. SE 0.2, $Z = 2.0 > 1.96$, reject. The HT p.10 example gives $Z = -2.83$ and rejects.

Usual mistake. Writing "accept H_0". Failing to reject isn't evidence for H_0 (HT p.4).

HW5 1.2. p-value and conclusion

1. $p = P[\text{statistic this extreme} \mid H_0]$ (HT p.13).
2. Upper tail $1 - 0.9641 = 0.0359$. Two-sided, so double it, $p = 0.0718$.
3. $0.072 > 0.05$, fail to reject. It always agrees with the critical-value rule.

Usual mistake. Forgetting to double for two-sided, or saying "7% chance H_0 is true" (HT p.14).

HW5 1.3. $\alpha = 0.01$, $n = 36$. Smallest $\bar{x} > 2.5$ that rejects

1. Two-sided, so $z[0.005] = 2.576$.
2. $\bar{x} > 2.5 + 2.576(0.1333) = 2.5 + 0.343 = 2.843$.
3. A one-sided $H_1 : \mu > 2.5$ would use $z[0.01] = 2.326$, giving 2.810.

Answer. About 2.84 h two-sided (2.81 one-sided). Matches main.tex.

Power version. Runs to detect $\delta = 0.2$ h with 80% power at 0.05 (HT p.19) are $((1.96 + 0.84) \cdot 0.8/0.2)^2 = 11.2^2 = 125.4$, so 126.

HW5 2.1. Suspect the mean time to first fault is below 740 h. H_0 and H_1?

1. The claim to establish goes in H_1, because a test only supports a claim by rejecting H_0 (HT p.4).
2. $H_0 : \mu = 740$ (or ≥ 740), $H_1 : \mu < 740$, one-sided (HT p.5).

Usual mistake. Putting the claim in H_0, so a small noisy experiment "supports" it by failing to reject. Picking the side after seeing the data is a disguised two-sided test at double α (HT p.5).

HW5 2.2. $n = 50$, $\bar{x} = 706$, $s = 96.1$, $\alpha = 0.01$. Can we confirm $\mu < 740$?

1. σ is unknown but $n = 50$, so s stands in and $t \approx Z$ (HT p.11).
2. SE $= 96.1/7.071 = 13.59$. $Z = -34/13.59 = -2.50$.
3. One-sided, reject if $Z < -2.326$. Since $-2.50 < -2.326$, reject.
4. $p = 1 - 0.9938 = 0.0062 < 0.01$.

Answer. Reject, so the data confirm $\mu < 740$ at 0.01. With t_{49} (crit 2.405) the verdict is the same.

Small version. $n = 36$, $\bar{x} = 716$, $s = 60$. SE 10, $Z = -2.4 < -2.326$, reject, $p = 0.0082$.

Exam variant, small n and the side matters. $n = 9$, $\bar{x} = 12$, $s = 3$, $\mu_0 = 10$, population roughly normal. SE $= 3/3 = 1$, $t = 2.0$ with 8 df. Two-sided at 0.05 needs 2.306, so fail to reject. One-sided $H_1 : \mu > 10$ (stated before the data) needs $t[0.05, 8] = 1.860$, so reject. That's why the side has to be fixed in advance (HT p.5, HT p.11).

HW5 Problem 3. Hypothesis testing in the wild, questions 0 to 6

Checklist.

0. Paper and why.
1. H_0, H_1, one or two-sided. Reconstruct if missing and say so.
2. Test and assumptions (normality, independence, pairing, equal variances). The unit, and does n count units or measurements (HT p.20)?
3. α, n, power. With no power analysis, the detectable effect is about 2.8×SE at $\alpha = 0.05$ and 80% power (HT p.19).
4. Count the tests. FWER $= 1 - (1 - \alpha)^m$ (HT p.27). Any correction, and FWER or FDR (HT p.28)?
5. Stopping rule (peeking, HT p.23 to 24), interference (HT p.22), right estimand.
6. Rewrite as an effect size with a CI (HT p.16, p.18). Big enough to matter?

His paper. Spang et al., Unbiased Experiments in Congested Networks, IMC 2021 (entry 8).

0. Closest to video streaming, and his comparisons share a bottleneck too.
1. Not stated. Reconstructed H_0: TTE $= 0$ against H_1: TTE $\neq 0$, two-sided since the CIs are two-sided.
2. No named test. Regression on hourly means with Newey-West SEs. The unit is the session but n counts hour cells, since sessions in one hour share congestion. An A/A test and a baseline week check that the links match.
3. $\alpha = 0.05$ implied. About 14M sessions but only a few dozen hour cells. No power analysis. SE about 2.6, so the detectable effect is about $2.8 \times 2.6 \approx 7\%$.
4. 28 intervals in Fig. 5 (7 metrics × 4), plus 24 baseline metrics. FWER $= 1 - 0.95^{28} = 0.76$. No correction, and FDR fits the claim better.
5. 5 days, not said to be fixed in advance. Interference is the paper's point. The naive A/B test measures the effect at 5% allocation, not the effect of deploying to everyone.
6. Capping raised throughput 12% (95% CI about 7% to 17%), while the naive test said −5% (about −8% to −2%). The claim gets stronger since the sign flips, and the exact size is less certain.

By hand. 10 tests at 0.05 with every H_0 true give FWER $= 1 - 0.95^{10} = 0.40$ (HT p.27). Bonferroni uses $0.05/10 = 0.005$ each (HT p.28).

Exam scenario. "A new congestion controller ran on 10% of flows sharing links with the old one. We checked p daily and stopped on day 6 at $p = 0.04$. Of 8 metrics, throughput was significantly better."

1. No H_0 stated. Reconstruct H_0 no throughput difference, H_1 better, and ask if the side was fixed in advance (HT p.5).
2. Unit is the flow, but flows on one link aren't independent (HT p.20).
3. No power analysis, n wasn't fixed (HT p.19).
4. 8 metrics give FWER $= 1 - 0.95^8 = 0.34$ with no correction (HT p.27 to 28).
5. Daily peeking inflates the Type I error well past 5% (HT p.23 to 24). At 10% allocation the new flows interfere with the old, so this isn't the effect of deploying to everyone (HT p.22).
6. Report the throughput difference with a CI, not "significantly better" (HT p.16).

HW6. Comparing two alternatives

HW6 P1. Same workload on routing A and B, 9 rows of delay. Is A better at 95%?

A	B	d	d^2	A	B	d	d^2
115.0	125.3	−10.3	106.09	120.3	87.9	32.4	1049.76
89.3	97.6	−8.3	68.89	98.9	104.2	−5.3	28.09
102.1	85.3	16.8	282.24	87.6	96.3	−8.7	75.69
94.3	101.3	−7.0	49.00	104.5	113.2	−8.7	75.69
117.3	112.7	4.6	21.16				

1. The same workload ran under both, so paired. Work on $d_i = A_i - B_i$, where negative favours A (Comp p.5, p.9).
2. $\sum d = 53.8 - 48.3 = 5.5$, $\bar{d} = 0.611$.
3. $\sum d^2 = 1756.61$. $s_d^2 = (1756.61 - 5.5^2/9)/8 = (1756.61 - 3.36)/8 = 219.16$, $s_d = 14.80$.
4. SE $= 14.80/3 = 4.935$. $t[0.025, 8] = 2.306$, half-width 11.38.
5. CI $= (-10.77, 11.99)$ contains 0.

Answer. You can't say A is better at 95%. Matches main.tex. It isn't evidence they're equal either, since the CI allows A 10.8 better or 12.0 worse.

Check. "Is A better than B?" is a one-sided question by HT p.5, and he used a two-sided interval. The verdict doesn't change. A is better only if $\bar{d} < 0$, and $\bar{d} = +0.611$ points the other way, so the one-sided test with $t[0.05, 8] = 1.860$ can't reject either.

By hand. Four differences 2, 4, 3, 3. $\bar{d} = 3$, $s_d^2 = 2/3$, $s_d = 0.816$, SE 0.408, and $3 \pm 3.182(0.408) = (1.70, 4.30)$ excludes 0. Sign test on the HW6 data (Comp p.23). 3 positive out of 9, $P[S \leq 3] = (1 + 9 + 36 + 84)/512 = 130/512$, two-sided $p = 0.51$.

Usual mistake. Analysing paired data as unpaired. Comp p.8 shows the same data giving $(0.65, 3.00)$ paired and $(-22.0, 25.7)$ unpaired. The df come from pairs, $n - 1 = 8$ (Comp p.9).

HW6 P2. CPU times, 30 on A and 24 on B. Is A faster at 90%?

1. Different transactions and unequal sizes, so unpaired. Welch by default (Comp p.12 to 13).
2. A has $n = 30$, $\bar{x} = 1.8830$, $s = 1.9017$. B has $n = 24$, $\bar{x} = 2.2623$, $s = 2.9108$. Difference −0.3793.
3. $s_A^2/n_A = 3.6166/30 = 0.12055$, $s_B^2/n_B = 8.4727/24 = 0.35303$, so $s = \sqrt{0.47358} = 0.6882$.
4. $\nu = 0.47358^2/(0.12055^2/29 + 0.35303^2/23)$, which is $0.22428/(0.000501 + 0.005419) = 37.89$.
5. $t[0.05, 37.89] = 1.686$, half-width 1.160. CI $= (-1.540, 0.781)$.

Answer. 0 is inside, so A isn't shown to be faster. Welch $t = -0.55$, $p = 0.58$. Matches main.tex. Rounding ν down to 37 by hand gives 1.687 and $(-1.540, 0.782)$.

Check. The same one-sided point applies. "Is A faster" is one-sided. At 90% one-sided, $t[0.10, 37.9] = 1.304$ and the upper bound is $-0.379 + 1.304(0.688) = 0.518 > 0$. Still not shown, so the verdict holds, but say the test is one-sided.

By hand. A $= 2, 4, 6, 8$ and B $= 6, 8, 10, 12, 14$. Means 5 and 10, variances 6.67 and 10. Pieces 1.667 and 2, $s = \sqrt{3.667} = 1.915$. $\nu = 13.44/(0.926 + 1) = 6.98$, round down to 6. The 90% CI is $-5 \pm 1.943(1.915) = (-8.72, -1.28)$, which excludes 0, so A is smaller.

Usual mistake. Using $n - 1$ of one sample as the df, or reading overlap of two separate CIs as "no difference" (Comp p.15 to 16).

Slide twin, paired (Comp p.11). Cache hit ratio on 6 workloads under policies A and B

1. Differences $+0.07, +0.09, -0.05, -0.06, +0.10, -0.03$. $\sum d = 0.12$, $\bar{d} = 0.020$.

2. $\sum d^2 = 0.0049 + 0.0081 + 0.0025 + 0.0036 + 0.0100 + 0.0009 = 0.0300$.
3. $s_d^2 = (0.0300 - 0.12^2/6)/5 = (0.0300 - 0.0024)/5 = 0.00552$, $s_d = 0.0743$.
4. SE $= 0.0743/\sqrt{6} = 0.0743/2.449 = 0.0303$.
5. The slide multiplies by $t[0.05, 5] = 2.015$, so its interval $(-0.041, +0.081)$ is a 90% interval. For 95% use $t[0.025, 5] = 2.571$, giving 0.020 ± 0.078.

Contains 0, so no difference shown, and the width is the finding (Comp p.11).

Slide twin, unpaired Welch (Comp p.14). A has $n = 7$, mean 0.2657, SD 0.0934. B has $n = 9$, mean 0.4011, SD 0.1447

1. Pieces $0.0934^2/7 = 0.001246$ and $0.1447^2/9 = 0.002326$, sum 0.003572, $s = 0.0598$.
2. $\nu = 0.003572^2/(0.001246^2/6 + 0.002326^2/8) = 13.6$.
3. $t[0.05, 13.6] = 1.765$, half-width 0.106, difference -0.135.
4. 90% CI $= (-0.241, -0.029)$ excludes 0, so B has the higher hit ratio, by 3 to 24 points.

HW6 Problem 3. Comparing two systems in the wild, 8 questions

Checklist.

1. Paper and why.
2. Design. Paired or unpaired, the unit, units per system (Comp p.5).
3. If unpaired, could it be paired? How much variance is between units (Comp p.6)?
4. Which test, and are df and an exact p reported? If no test, what instead?
5. Is the response normal, or a bounded score, ratio, quantile? Would a rank test or bootstrap be better (Comp p.17 to 18)?
6. Are error bars defined in the caption? Any conclusion from overlap (Comp p.15 to 16)?
7. Effect size in real units with an interval. Big enough, and who decides (Comp p.24)?
8. Rewrite the headline with an interval plus what it couldn't detect (Comp p.25).

His paper. Miller, Adding Error Bars to Evals, arXiv 2411.00640, 2024 (entry 3).

1. He compares VLMs on the same benchmark questions.
2. Paired, with the question as the unit (the cluster for MGSM). Example models on 5000 MATH, 164 HumanEval, 2500 MGSM.
3. Already paired. In Miller's example a correlation of 0.5 cuts the variance from 1/6 to 1/9, so the SE shrinks by $\sqrt{2/3} = 0.82$, about 18%.
4. Paired z on per-question differences, ± 1.96 SE. No df, no p. Significant if the CI excludes 0.
5. Scores are 0 or 1, not normal, so it leans on the CLT. Fine for 5000, but a paired bootstrap or sign test is safer for 164.
6. No figures, only SEs in tables. Single-model CIs on MATH overlap though the paired CI excludes 0.
7. MATH difference +2.5 points, 95% CI (1.2, 3.8). Whether that matters is left to the user.
8. Galleon beat Dreadnought by 2.5 points on MATH (1.2 to 3.8). HumanEval -3.1 (-7.2, 1.0) and MGSM -2.7 (-6.1, 0.7) are consistent with no difference. HumanEval's SE is 2.1, so it couldn't reliably detect under about $2.8 \times 2.1 \approx 6$ points.

Exam scenario. "Compiler X and Y were each run on the same 8 benchmarks. An unpaired t test gave $p = 0.4$, and the error bars overlap, so the compilers perform the same."

2. Same benchmarks under both, so it's paired with the benchmark as the unit (Comp p.5).
3. Unpaired throws away the correlation. Between-benchmark spread swamps the difference, as in Comp p.8 where the same data give $p = 0.87$ unpaired and $p = 0.008$ paired.
4. Report t with df and the exact p (Comp p.26).
5. Run times are skewed, so analyse log times or use the signed-rank test (Comp p.18, Comp p.21).
6. The bars aren't defined, and overlap isn't a test (Comp p.15).
7. Give the mean difference with a CI. "Perform the same" doesn't follow from a non-significant result.

One checklist for any experiment on the exam

1. Claim and hypotheses, with the direction chosen before the data (HT p.4 to 5).
2. Unit and n. Count runs, not requests from one run (HT p.20, CI p.66).
3. Paired or unpaired, and could it be paired (Comp p.5)?
4. Test fits the response, whether z, t, Welch, rank or bootstrap (Comp p.17 to 18, CI p.67).
5. Power or detectable effect (HT p.19).
6. Number of tests, and peeking (HT p.23 to 28).
7. Effect size with an interval in real units (HT p.16, Comp p.24).
8. What the interval misses, such as systematic error and interference (CI p.17, HT p.22).
9. Reading. No "equivalent" from non-significance, no verdicts from overlap, no "95% probability" for a CI (HT p.14, Comp p.15, CI p.33).

Practice. "Our scheduler cuts mean latency from 12.1 to 11.4 ms, $p < 0.05$, from one run of each with 10,000 requests, with the best of three workloads shown." The unit is the run, so $n = 1$ per system and the p means nothing. Three workloads are a hidden family of tests. There's no interval on the 0.7 ms. Fix it with several independent runs per system on all three workloads, paired by workload, reporting the difference and its CI.

Homework-style mock problems, no calculator

Mock 1 (HW3 style). Three runs of equal work report 40, 60 and 120 req/s. Mean rate?

1. Rates with equal work, so harmonic (Summ p.28).
2. $1/40 + 1/60 + 1/120 = 0.025 + 0.0167 + 0.0083 = 0.05$.
3. $x_H = 3/0.05 = 60$ req/s. The AM would say 73.3.

Mock 2 (HW4 style). $n = 25$, $\bar{x} = 50$, $s = 10$, σ unknown. 90% CI?

1. $n < 30$ and σ unknown, so t with 24 df, assuming roughly normal data (CI p.37).
2. SE $= 10/5 = 2$. $t[0.05, 24] = 1.711$, half-width 3.42.
3. CI = (46.58, 53.42).

Mock 3 (HW5 style). $\sigma = 4$ known, $n = 64$, $\bar{x} = 51$, $H_0 : \mu = 50$, two-sided, $\alpha = 0.05$

1. SE $= 4/8 = 0.5$, $Z = 1/0.5 = 2.0$.
2. $|Z| > 1.96$, reject. $p = 2(1 - 0.9772) = 0.0456$.
3. At $\alpha = 0.01$ (2.576) you'd fail to reject. Report p and the CI 51 ± 0.98, not just the verdict (HT p.16).

Mock 4 (HW6 style). Paired differences 1, 3, 2, 4, 0 on 5 benchmarks. 95% CI on the mean difference

1. $\bar{d} = 10/5 = 2$. Deviations $-1, 1, 0, 2, -2$, squares sum to 10, $s_d^2 = 10/4 = 2.5$, $s_d = 1.581$.
2. SE $= 1.581/\sqrt{5} = 0.707$. $t[0.025, 4] = 2.776$, half-width 1.963.
3. CI = (0.04, 3.96), just excludes 0, so a difference is shown. The size is anywhere from almost nothing to about 4, which you should say.

Practice midterm

Slides: Intro, Metrics, Advisor, Summ, CS701, CI, Pub, Ethics, HT, Comp, Meas, plus the study guides and in-the-wild handouts for those lectures. Homework: the problems copy the style of HW3 to HW6, with fresh numbers. This is a full-length mock exam. Set a timer for two hours and put your phone and laptop away, because the real exam is open book and open notes but allows no electronics. Work with a pencil, your notes and the table values below, and write each answer down before you open its solution. A fair pace is about 30 minutes for Part A, 75 minutes for Part B and 15 minutes for Part C. Every solution shows each arithmetic step and cites the slide it comes from, and every number was computed twice.

These are the only critical values the exam uses. They're the values a printed t and normal table gives, so read them off your own table and check they match. The brackets are the slides' upper-tail notation, so $t[0.025, 7]$ is the t value with area 0.025 above it on 7 degrees of freedom (CI p.37, HT p.5). $\Phi(z)$ is the area below z in a normal table.

Upper-tail area	0.10	0.05	0.025	0.01	0.005
z	1.282	1.645	1.960	2.326	2.576
t, 5 df	1.476	2.015	2.571	3.365	4.032
t, 7 df	1.415	1.895	2.365	2.998	3.499
t, 8 df	1.397	1.860	2.306	2.896	3.355
t, 15 df	1.341	1.753	2.131	2.602	2.947

One normal-table value is also needed, $\Phi(2.5) = 0.9938$.

Part A. Concepts (17 short questions, about 30 minutes)

Answer each in two or three sentences. For true or false items, say which and give the reason, since the reason is where the marks are.

A1. True or false, with a reason. (a) If MIPS(A) > MIPS(B), machine A runs your program faster than machine B. (b) Linearity is one of the properties a good metric must have. (c) Is MFLOPS a means-based or an ends-based metric?

A2. A web search returns 25 pages. 15 of them are relevant, and it misses 35 other relevant pages. What are its precision and recall?

A3. Feitelson gives three meanings of "experimental computer science". Name them and say which one CS 700 is about.

A4. A first-year student says, "The advisor form isn't due until I've taken 24 credits, so I'll start looking that semester, and I'll open my first meeting by asking the professor to be my advisor." Give two reasons this plan is bad.

A5. True or false, with a reason. (a) Your CS 701 supervisor automatically becomes your PhD advisor. (b) If you can't find a CS 701 project this spring, you can take it next year. (c) If no faculty member agrees, the department will assign you one. (d) You should wait for one professor's answer before contacting a second.

A6. Which of these is acceptable for CS 701, and why? (i) A survey of 30 papers on federated learning. (ii) Reimplementing three published cache policies and evaluating them with a new metric. (iii) A deliverable that reads "paper accepted at a conference by May". (iv) A deliverable that reads "a draft paper to be revised and submitted for publication later".

A7. An author's papers have 15, 12, 7, 5, 5, 3 and 1 citations. What is the h-index?

A8. (a) A conference uses double-blind review. What must your submission do, and what happens if it doesn't? (b) True or false. It's fine to send the same paper to two conferences at once as long as you withdraw one when the other accepts.

A9. Classify each as fabrication, falsification, plagiarism or none. (a) You drop two runs that went against your trend and don't mention it. (b) You run out of time and type in plausible numbers for one benchmark. (c) You paste a sentence from a survey into your related work with no quotation marks or citation. (d) After publishing, you find a timer bug that inflated your speedup by 3%.

A10. (a) You used ChatGPT to draft part of the related work in a paper for an ACM venue. Can you list it as a co-author, and what must you do? (b) You're extending your IEEE conference paper into a journal paper. What does the journal version need?

A11. Which index of dispersion would you report for (a) CPU utilization in percent, (b) an unbounded, unimodal, symmetric quantity, (c) right-skewed request latencies with a few huge outliers?

A12. From one sample you get a 95% confidence interval of (47.6, 52.4) ms for the mean run time. Which statements are correct? (i) There's a 95% probability that μ lies in (47.6, 52.4). (ii) If you repeated the whole experiment many times, about 95% of the intervals built this way would contain μ. (iii) About 95% of individual runs take between 47.6 and 52.4 ms.

A13. True or false. Running a benchmark 1000 times instead of 10 makes the result more accurate.

A14. A paper compares two schedulers with 5 runs each, gets p = 0.30, and concludes that "the two schedulers perform the same." (a) What's wrong? (b) If the schedulers really differ, what kind of error did the paper make, and what is its probability called? (c) At a fixed n, what happens to that probability if you lower α?

A15. You test 20 benchmarks, each at $\alpha = 0.05$, and no real effect exists on any of them. (a) How many "significant" results do you expect? (b) What's the formula for the chance of at least one, and its value from the slides? (c) What per-test level would Bonferroni use?

A16. True or false. The 95% intervals for systems A and B overlap a little, so there's no significant difference between them.

A17. (a) Pick counting, tracing or sampling for each, with a reason. (i) Total L2 misses in a 60-second run. (ii) The code path behind the one-in-a-million request that takes 100 times the median. (iii) A two-hour job where you have no idea where the time goes. (b) True or false. 20 iterations inside one process give you n = 20 independent measurements.

Part B. Problems (8 problems, about 75 minutes)

Show every step, since that's how the homework was graded. State which formula you use and why before you plug in.

B1. Choosing a mean. For each part, say which mean (arithmetic, harmonic or geometric) you'd use and why, then compute it. (a) Five runs of a build take 120, 80, 150, 90 and 160 ms. What's the mean build time? (b) The same 120 MB file is sent over three paths at 40, 60 and 120 MB/s. What's the mean transfer rate? (c) Three features are added one on top of the other, and each improves solution quality over the previous version by 8%, 25% and 28%. What's the mean improvement per feature? You may use $1.2^3 = 1.728$.

B2. Quartiles. Twelve request latencies in ms, in the order measured.

23	15	60	18	31	12	25	20	14	28	22	17

Using the course convention $f_i = (i - 0.5)/n$ with linear interpolation, find (a) the five-number summary, (b) the IQR and SIQR, (c) the 90th percentile. (d) Which center and which index of dispersion would you report, and why?

B3. Confidence intervals for a mean. (a) Eight independent runs of a compile job take these times in seconds.

52	46	50	54	50	46	52	50

Find the mean, the sample standard deviation, and the 90%, 95% and 99% confidence intervals for the mean. (b) Treating these 8 runs as a pilot, how many runs would you need to be 95% confident the mean is within ±2% of the true mean? (c) A different job is run 64 times, with $\bar{x} = 120$ ms and $s = 16$ ms. Give the 95% and 99% intervals, then test $H_0 : \mu = 125$ against a two-sided alternative at $\alpha = 0.05$ and $\alpha = 0.01$, with the p-value. (d) If that job had been run only 16 times with the same $\bar{x}$ and s, what would the 95% half-width be, and why is it so much wider?

B4. A proportion from a sampling profiler. A profiler interrupts every 10 ms during a 4-second run, which gives 400 samples. In 80 of them the program counter is inside `parse()`. (a) Find 90% and 95% confidence intervals for the fraction of time spent in `parse()`, and check that the normal approximation is reasonable. (b) How many samples do you need to be 95% confident that the estimate is within ±5% of the true fraction (relative error, as on the slides), and how long must the run be? (c) How would the answer to (b) change if the question asked for ±5 percentage points instead?

B5. Hypothesis test for a mean. A team believes its new memory allocator brings the mean allocation latency below the old allocator's long-run mean of 40 µs. They measure 16 independent runs, each in a fresh process, and get $\bar{x} = 37$ µs with $s = 6$ µs. Assume the run means are roughly normal. (a) State H_0 and H_1 and justify the choice. (b) Test at $\alpha = 0.05$. (c) Bracket the p-value using the table. (d) What's the verdict at $\alpha = 0.01$? (e) At $\alpha = 0.01$, what is the largest sample mean that would let you reject H_0? (f) A colleague reruns the analysis, gets p = 0.064 and calls the result not significant. What did they do differently, and who is right?

B6. Paired comparison. Two routing algorithms are run on the same six injected workloads, and the average end-to-end delay (ms) is recorded for each. Can you say with 95% confidence that algorithm A gives lower delay than B?

Workload	**1**	**2**	**3**	**4**	**5**	**6**
A	42	35	51	28	60	47
B	41	37	54	31	65	53

B7. Unpaired comparison with Welch. CPU time (ms) is measured for different random transactions on two systems, 6 on A and 8 on B. Is System B faster than System A at the 90% confidence level?

A	52	57	46	54	48	55		
B	45	41	47	44	50	42	46	45

B8. Measurement. (a) Your clock has a resolution of 1 µs and the operation you want to time takes about 2.6 µs. What can one reading return, and how big is the error? (b) Instead you time 10,000 operations in a loop and get 27,412 µs, and an empty loop of 10,000 iterations takes 1,412 µs. What's the time per operation, what's the resolution of that figure, and what assumption does it need? (c) A clock read costs 20 ns, and you bracket each operation with two reads. What fraction of the measured time is timer overhead if the operation takes 60 ns? If it takes 4 µs? (d) A benchmark runs in 80 s untraced and 86 s with two trace probes on. Alone, the probes cost 2.5 s and 2.0 s. Report the perturbation and explain why 2.5 + 2.0 isn't 6. (e) The variance between processes is $\sigma_p^2 = 48$ ms^2 and between iterations inside a process is $\sigma_i^2 = 16$ ms^2. With a budget of 16 measurements, find the standard error of the overall mean for 1 process × 16 iterations, 2 × 8, and 16 × 1.

Part C. Paper audit (1 question, about 15 minutes)

This is the in-class version of Problem 3 from HW4 to HW6. Read the made-up results paragraph and answer the questions under it, two or three sentences each, except C3, which needs arithmetic.

"We compare FastKV with RocksDB on a single 16-core server. For each system we ran YCSB workload A five times and report the average throughput. The first 30 seconds of each run were discarded as warm-up, and throughput was timed with gettimeofday(). All FastKV runs were done on Monday, and the RocksDB runs on Tuesday after a kernel update. FastKV reaches 118.2 Kops/s against 104.6 Kops/s for RocksDB, a 13% improvement. Figure 4 shows both means with error bars of ±1 standard deviation (2.0 for FastKV and 4.0 for RocksDB). Because the bars do not overlap, the improvement is statistically significant. We repeated the comparison on 12 more workloads, and FastKV was significantly faster ($p < 0.05$) on 4 of them. Averaged over all 13 workloads, FastKV's arithmetic mean throughput is 9% higher. Finally, the two systems have equivalent p99 latency (t test, $p = 0.41$)."

C1. What is the experimental unit, what is n, and is the headline design paired or unpaired? Could any part of it have been paired? C2. Is the error-bar argument valid? C3. Write H_0 and H_1 for the headline claim, then rewrite the claim as a 95% confidence interval for the difference using Welch's method. C4. Is the arithmetic mean the right summary across workloads? C5. What's wrong with "significantly faster on 4 of 12 more workloads"? C6. Name two problems with the measurement setup that no interval can catch. C7. What's wrong with the p99 sentence?

Practice midterm, solutions

Part A

A1. (a) False. MIPS counts instructions per second, and machines differ in how much work one instruction does, so a machine with more MIPS can still take longer on your program. That's exactly the slides' example of an unreliable metric (Metrics p.8, Metrics p.17).

(b) False. Linearity and ease of use are nice but not necessary. The required four are reliable, repeatable, consistent and independent (Metrics p.13).

(c) Means-based. It counts floating-point operations whether or not they did useful work, and it reports 0 for an integer program that still sorts or searches. On the slides' spectrum MFLOPS sits near the means-based end with clock rate and MIPS, while execution time is the ends-based metric because it only counts what actually got done (Metrics p.19, Metrics p.27-28).

A2. Precision is the share of returned pages that are relevant, so it's $15/25 = 0.6$. Recall is the share of all relevant pages that were returned. There are $15 + 35 = 50$ relevant pages in total, so recall is $15/50 = 0.3$. This is the same setup as the slides' example with 30 returned, 20 relevant and 40 missed (Metrics p.32).

A3. The three are building systems as the counterpart to theory, experimentation as the feedback step in an engineering loop, and evaluating computer systems with the rigorous methods of the natural sciences. CS 700 is about the third one, which is why the course spends its time on measurement, error and statistics (Intro p.10).

A4. First, the 24-credit point is a hard deadline, and most faculty want an informal preliminary research co-operation phase before they agree to advise anyone. That phase can take a semester or more, so starting in the last semester leaves no room for it (Advisor p.3, Advisor p.7-8).

Second, the slides say not to open with "can you be my PhD advisor?". You should explain your background and interests and ask how you can get involved in their research, because the faculty member has to see you work first and may already be over-committed (Advisor p.11-12).

A5. (a) False. Declaring the PhD advisor is a separate form and a separate process within the first 24 credits, and many students end up with a different advisor and area than their 701 project (CS701 p.16, CS701 p.18).

(b) False. The catalog rule doesn't allow it, and 701 is only offered in Spring (CS701 p.25, CS701 p.2).

(c) False. A faculty member has to agree to supervise, the department doesn't assign one (CS701 p.26).

(d) False. You should contact several faculty at once, because answers take time and the form is due Oct 31 (CS701 p.21, CS701 p.7).

A6. (i) Not acceptable on its own. A literature review is often part of a 701 project, but the project can't consist only of one, because it must have a hands-on component such as an implementation or experiments (CS701 p.10-11).

(ii) Acceptable. Implementing techniques from several papers and evaluating them, possibly with new metrics, is one of the slides' examples (CS701 p.12).

(iii) Not acceptable, because a conference or journal submission completed by the end of Spring can't be a deliverable (CS701 p.15).

(iv) Acceptable. The slides give this exact wording as the allowed alternative, and the report can then take the form of the draft paper (CS701 p.15).

A7. The h-index is the largest h such that h papers each have at least h citations. Five papers have 5 or more citations (15, 12, 7, 5, 5), so $h \geq 5$. Only three papers have 6 or more, so $h = 6$ fails. The h-index is 5 (Pub p.6).

A8. (a) You must leave out your names and affiliations and avoid indirect hints about who you are. A submission that breaks the double-blind rules is rejected right away, without review. The point is that reviewers judge the paper and not the authors' reputation (Pub p.21).

(b) False. A paper under review somewhere can't be submitted elsewhere, and the same holds for papers with substantial overlap. Dual submissions are often caught and draw severe sanctions (Pub p.20, Ethics p.14).

A9. (a) Falsification, because omitting data so the record misrepresents the research is the definition (Ethics p.13, Ethics p.20).

(b) Fabrication, because it means making up results and reporting them (Ethics p.20).

(c) Plagiarism. Verbatim text needs quotation marks and a citation, and even a paraphrase needs the citation (Ethics p.8-10).

(d) None. Research misconduct excludes honest error, though you should still correct the record (Ethics p.18).

A10. (a) No. ACM doesn't allow generative AI tools as authors. Using them to create content is allowed but must be fully disclosed, for example in the acknowledgements. Plain spelling, grammar and translation tools are the exception and don't need disclosure (Ethics p.24). You also take full responsibility for anything the tool produced, including plagiarism or fabricated facts (Ethics p.22).

(b) It needs substantially more material, typically 30% more for IEEE, and it must cite and attach the conference version with an explanation of the differences (Ethics p.14). Outside that case, copying your own published text counts as self-plagiarism (Ethics p.12).

A11. (a) The range, because the quantity is bounded between 0 and 100 (Summ p.49).

(b) The coefficient of variation $s/\bar{x}$, because for a unimodal symmetric distribution the standard deviation describes the spread well and dividing by the mean makes it unit-free (Summ p.41, Summ p.49).

(c) Percentiles or the SIQR. Outliers inflate the standard deviation the same way they drag the mean, while quartiles depend only on order (Summ p.45, Summ p.49). For the same reason you'd pair it with the median instead of the mean (Summ p.18).

A12. Only (ii) is correct. The 95% belongs to the procedure. Out of 100 samples, about 95 of the intervals contain μ (CI p.33).

(i) is wrong because μ is a fixed number and this interval is fixed once computed, so it either contains μ or it doesn't (confidence-intervals study guide).

(iii) is wrong because the interval is for the mean, not for single runs. Its width shrinks like $1/\sqrt{n}$, so with enough runs it gets far narrower than the spread of individual runs (CI p.39).

A13. False. More runs narrow the confidence interval, and that's precision, which comes from random error. Accuracy depends on

systematic error, such as clock drift or forgetting to clear the cache, and repetition doesn't touch it (CI p.7, CI p.17). You just get a tighter interval around a possibly wrong number (Meas p.20).

A14. (a) Failing to reject H_0 isn't evidence for H_0. With n = 5 the test had little chance of seeing a difference, so absence of evidence isn't evidence of absence (HT p.4, HT p.14). The paper should report the interval for the difference, which would show how wide the plausible range is (HT p.16).

(b) A Type II error, failing to reject a false H_0. Its probability is β, and $1-\beta$ is the power (HT p.6).

(c) β goes up. At a fixed n, lowering α raises β, and the only way to lower both is to collect more data (HT p.6).

A15. (a) Each test has a 0.05 chance of a false positive, so you expect $20 \times 0.05 = 1$ of them.

(b) For independent tests the family-wise error rate is $1-(1-\alpha)^m = 1-0.95^{20}$, which the slides give as 64% (HT p.27).

(c) Test each at $\alpha/m = 0.05/20 = 0.0025$, which keeps the family-wise rate at or below 0.05 under any dependence (HT p.28).

A16. False. Non-overlap implies a difference, but overlap implies nothing. For 95% intervals, an overlap of up to about half a margin of error can still be significant at 0.05, because the standard error of the difference is smaller than the sum of the two half-widths. The right move is an interval on the difference (Comp p.15-16).

A17. (a)(i) Counting. You need only an exact total, and a counter gives it with tiny overhead (Meas p.7-8).

(a)(ii) Tracing. You need context about a rare event, and sampling would almost surely miss it (Meas p.8).

(a)(iii) Sampling. Its overhead is set by your sampling rate and not by the workload, which matters for a long run you can't afford to slow down (Meas p.7).

(b) False. Repetition has nested levels, and everything that varies between processes, like memory layout or JIT decisions, is frozen across those 20 iterations. n counts independent experimental units, here processes, not measurements (Meas p.25, HT p.20).

Part B

B1. Choosing a mean.

(a) Arithmetic mean, because these are times and the sum of the times is the total time, so the mean should be directly proportional to it (Summ p.25, Summ p.27).

$$\bar{x}_A = \frac{120+80+150+90+160}{5} = \frac{600}{5} = 120 \text{ ms}.$$

(b) Harmonic mean, because these are rates for the same amount of work, and a mean rate has to be inversely proportional to the total time (Summ p.28, Summ p.32).

$$\bar{x}_H = \frac{3}{\frac{1}{40}+\frac{1}{60}+\frac{1}{120}} = \frac{3}{\frac{3+2+1}{120}} = \frac{3}{6/120} = \frac{3\times 120}{6} = 60 \text{ MB/s}.$$

Check it the long way. The transfers take $120/40 = 3$ s, $120/60 = 2$ s and $120/120 = 1$ s, so 360 MB move in 6 s, which is $360/6 = 60$ MB/s. The arithmetic mean $(40+60+120)/3 = 73.3$ MB/s would overstate the real rate because it ignores that the slow path takes the most time.

(c) Geometric mean, because each feature builds on the previous one, so the improvements multiply (Summ p.30). Turn each percentage into a ratio and multiply.

$$1.08 \times 1.25 = 1.35, \qquad 1.35 \times 1.28 = 1.728, \qquad \bar{x}_G = (1.728)^{1/3} = 1.2.$$

So the mean improvement is 20% per feature. Check it by compounding, $1.2^3 = 1.728$, which is the real total gain of 72.8%. The arithmetic mean of 8, 25 and 28 is 20.33%, and compounding that gives $1.2033^3 \approx 1.742$, which overshoots the true total. This is HW3 Problem 1 with smaller numbers.

B2. Quartiles.

Sort first. The ordered values $x_{(1)}, \ldots, x_{(12)}$ are

i	1	2	3	4	5	6	7	8	9	10	11	12
$x_{(i)}$	12	14	15	17	18	20	22	23	25	28	31	60

The quantile function puts $x_{(i)}$ at $p = (i-0.5)/n$ and interpolates in between (Summ p.59). Solving $(i-0.5)/n = p$ gives the position $r = np + 0.5$. With $n = 12$, take the whole part j of r and the fraction $r-j$, then $Q(p) = x_{(j)} + (r-j)\big(x_{(j+1)} - x_{(j)}\big)$.

(a) The quartiles.

$$Q_1:\ r = 12(0.25)+0.5 = 3.5, \quad Q_1 = 15 + 0.5(17-15) = 16.$$

$$Q_2:\ r = 12(0.5)+0.5 = 6.5, \quad Q_2 = 20 + 0.5(22-20) = 21.$$

$$Q_3:\ r = 12(0.75)+0.5 = 9.5, \quad Q_3 = 25 + 0.5(28-25) = 26.5.$$

The five-number summary is min 12, $Q_1 = 16$, median 21, $Q_3 = 26.5$, max 60 (Summ p.50).

(b) IQR $= Q_3 - Q_1 = 26.5 - 16 = 10.5$ ms and SIQR $= 10.5/2 = 5.25$ ms (Summ p.45).

(c) $r = 12(0.9)+0.5 = 11.3$, so $Q(0.9) = x_{(11)} + 0.3\big(x_{(12)} - x_{(11)}\big) = 31 + 0.3(29) = 31 + 8.7 = 39.7$ ms.

(d) The sum is 285, so the mean is $285/12 = 23.75$ ms, which sits above 8 of the 12 values because the single 60 pulls it up. Report the median of 21 ms and the SIQR of 5.25 ms, because the data are skewed with an outlier and both are based on order, so the outlier can't move them (Summ p.18, Summ p.45, Summ p.49).

Excel's PERCENTILE interpolates at position $1+(n-1)p$ instead and would give $Q_1 = 16.5$ and $Q_3 = 25.75$. Use the course convention on the exam, since that's what HW3 Problem 2 used.

B3. Confidence intervals for a mean.

(a) The sum is $52+46+50+54+50+46+52+50=400$, so $\bar{x}=400/8=50$ s. The deviations from 50 are 2, -4, 0, 4, 0, -4, 2, 0 and their squares are 4, 16, 0, 16, 0, 16, 4, 0.

$$s^2=\frac{\sum(x_i-\bar{x})^2}{n-1}=\frac{56}{7}=8, \qquad s=\sqrt{8}\approx 2.828 \text{ s}, \qquad \frac{s}{\sqrt{n}}=\sqrt{\frac{8}{8}}=1.$$

With $n=8$ and σ unknown you need the t distribution with $n-1=7$ degrees of freedom, assuming the run times are roughly normal (CI p.27, CI p.37). The interval is $\bar{x}\pm t[\alpha/2,\,7]\,s/\sqrt{n}$.

$$90\%:\ 50\pm 1.895(1)=(48.105,\ 51.895) \text{ s}$$

$$95\%:\ 50\pm 2.365(1)=(47.635,\ 52.365) \text{ s}$$

$$99\%:\ 50\pm 3.499(1)=(46.501,\ 53.499) \text{ s}$$

Higher confidence means a wider interval, because you need to cover more of the t distribution (CI p.32).

(b) The slides set the half-width equal to a fraction e of the mean and solve for n, using z (CI p.40).

$$n=\left(\frac{z[0.025]\,s}{\bar{x}\,e}\right)^2=\left(\frac{1.96\times 2.828}{50\times 0.02}\right)^2=\left(\frac{5.543}{1}\right)^2\approx 30.73.$$

A cleaner way by hand is $n=1.96^2\times s^2/(\bar{x}e)^2=3.8416\times 8/1=30.73$. Round up to 31 runs. The pilot gives s, which is the slides' way around needing s before you have data (CI p.41). Read the wording carefully, though. On CI p.42-43 "within 7%" was turned into $\pm 3.5\%$, so check whether a question gives the total width or the half-width.

(c) With $n=64>30$, s is a good stand-in for σ and the normal distribution is fine (CI p.27, CI p.31). The standard error is $16/\sqrt{64}=16/8=2$ ms.

$$95\%:\ 120\pm 1.960(2)=120\pm 3.92=(116.08,\ 123.92) \text{ ms}$$

$$99\%:\ 120\pm 2.576(2)=120\pm 5.152=(114.848,\ 125.152) \text{ ms}$$

For the test, $H_0:\mu=125$ and $H_1:\mu\neq 125$.

$$Z=\frac{\bar{x}-\mu_0}{s/\sqrt{n}}=\frac{120-125}{2}=-2.5.$$

At $\alpha=0.05$, $|Z|=2.5>1.960$, so reject H_0. At $\alpha=0.01$, $|Z|=2.5<2.576$, so don't reject. The two-sided p-value is $2\big(1-\Phi(2.5)\big)=2(1-0.9938)=2(0.0062)=0.0124$, which sits between 0.01 and 0.05 and gives the same two verdicts. The intervals agree too, since 125 is outside the 95% interval and inside the 99% one. That's the slides' point that the non-rejection region is the confidence interval (HT p.8, HT p.13).

(d) Now $n=16<30$, so use $t[0.025,\,15]=2.131$, and the standard error is $16/\sqrt{16}=4$ ms. The half-width is $2.131\times 4=8.524$ ms, against 3.92 ms before, about 2.2 times wider. Two things did it. The standard error doubled because $\sqrt{64}/\sqrt{16}=2$, and the critical value grew from 1.960 to 2.131 because t has heavier tails to pay for estimating σ from a small sample (CI p.35, CI p.39). This is HW4 Problem 1 part 4.

B4. A proportion from a sampling profiler.

(a) A sampling profiler reports a proportion, so the proportion interval applies (CI p.51-53, measurement study guide).

$$p=\frac{80}{400}=0.2, \qquad \sqrt{\frac{p(1-p)}{n}}=\sqrt{\frac{0.2\times 0.8}{400}}=\sqrt{\frac{0.16}{400}}=\sqrt{0.0004}=0.02.$$

The interval is $p\pm z_{\alpha/2}\sqrt{p(1-p)/n}$ (CI p.48).

$$90\%:\ 0.2\pm 1.645(0.02)=0.2\pm 0.0329=(0.1671,\ 0.2329)$$

$$95\%:\ 0.2\pm 1.960(0.02)=0.2\pm 0.0392=(0.1608,\ 0.2392)$$

The check is that $np=80$ and $n(1-p)=320$ are both well above 10, so the normal approximation to the binomial is fine (CI p.47, confidence-intervals study guide). If either were small, the Wald interval would be unreliable and you'd switch to Wilson or Agresti-Coull (CI p.49). So the profiler's "20%" really means somewhere between about 16% and 24%.

(b) The slides write the half-width as a fraction e of p, so $ep=0.05\times 0.2=0.01$ (CI p.54).

$$n=\frac{z_{\alpha/2}^2\,p(1-p)}{(ep)^2}=\frac{1.96^2\times 0.2\times 0.8}{(0.01)^2}=\frac{3.8416\times 0.16}{0.0001}=\frac{0.614656}{0.0001}=6146.56.$$

Round up to 6147 samples. At one sample every 10 ms the run must last $6147\times 0.01=61.47$ s, about a minute. This is the same calculation as the slides' OS-time example, which needed 3.46 hours for $\pm 0.5\%$ relative error (CI p.55-57).

(c) "± 5 percentage points" is an absolute half-width of 0.05, so the denominator becomes $0.05^2=0.0025$ and $n=0.614656/0.0025=245.9$, which rounds up to 246. The relative version needs 25 times more samples here, because $0.05/0.01=5$ and n grows with the square. The measurement study guide uses the absolute form, so always check which one the question means.

B5. Hypothesis test for a mean.

(a) The team wants to establish that the mean is below 40, so its negation goes in the null. $H_0:\mu=40$ (meaning $\mu\geq 40$) and $H_1:\mu<40$. If the test rejects H_0, that's evidence for the claim, while failing to reject would prove nothing (HT p.4). The direction was stated before the data were collected, so a one-sided test is legitimate (HT p.5). This mirrors HW5 Problem 2.

(b) σ is unknown and $n = 16$ is small, so use t with $n - 1 = 15$ degrees of freedom (HT p.11).

$$\frac{s}{\sqrt{n}} = \frac{6}{\sqrt{16}} = \frac{6}{4} = 1.5, \qquad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{37 - 40}{1.5} = -2.0.$$

For a one-sided test in the lower direction, reject if $t < -t[0.05, 15] = -1.753$. Since $-2.0 < -1.753$, reject H_0. At the 5% level the data support a mean below 40 µs.

(c) On the 15 df row, 2.0 falls between $t[0.05, 15] = 1.753$ and $t[0.025, 15] = 2.131$. So the one-sided p-value is between 0.025 and 0.05 (software gives 0.032). The p-value is the chance of a statistic at least this extreme if H_0 is true (HT p.13).

(d) Now the cutoff is $-t[0.01, 15] = -2.602$, and -2.0 isn't beyond it, so don't reject. That matches (c), since $p > 0.025 > 0.01$.

(e) Rejection needs $(\bar{x} - 40)/1.5 \le -2.602$, so

$$\bar{x} \le 40 - 2.602 \times 1.5 = 40 - 3.903 = 36.097 \text{ µs}.$$

Any sample mean of about 36.1 µs or less would reject at the 1% level. This is HW5 Problem 1 part 3 with t instead of z.

(f) The colleague used a two-sided test, which doubles the one-sided p-value, so their p is between 0.05 and 0.10 (software gives 0.064). The team is right here, because the direction was fixed by the claim before any data existed. The trap runs the other way. Picking one-sided after seeing which way the data went is a two-sided test at double the α in disguise (HT p.5). Whichever test you use, also report the effect, a drop of about 3 µs, with its interval (HT p.16, HT p.18). The one-sided 95% upper bound is $37 + 1.753 \times 1.5 = 39.63$ µs, which is below 40, matching the rejection in (b).

B6. Paired comparison.

Each workload is measured under both algorithms, so the data are paired and you analyze the six differences (Comp p.5). Take $d_i = B_i - A_i$, so a positive d favors A.

Workload	**1**	**2**	**3**	**4**	**5**	**6**
$d = B - A$	-1	2	3	3	5	6
$d - \bar{d}$	-4	-1	0	0	2	3
$(d - \bar{d})^2$	16	1	0	0	4	9

$$\bar{d} = \frac{-1 + 2 + 3 + 3 + 5 + 6}{6} = \frac{18}{6} = 3, \qquad s_d^2 = \frac{30}{5} = 6, \qquad \frac{s_d}{\sqrt{n}} = \sqrt{\frac{6}{6}} = 1.$$

The paired interval is a one-sample t interval on the differences with $n - 1 = 5$ degrees of freedom, since df counts pairs (Comp p.9).

$$\bar{d} \pm t[0.025, 5]\,\frac{s_d}{\sqrt{n}} = 3 \pm 2.571(1) = (0.429,\ 5.571) \text{ ms}.$$

The interval excludes 0 and lies entirely above it, so yes, with 95% confidence A has lower delay than B, by somewhere between 0.43 and 5.57 ms. The mean delay under A is $263/6 \approx 43.8$ ms, so that's roughly a 7% improvement with a wide range around it.

As a test, $H_0 : \mu_d = 0$ and $H_1 : \mu_d > 0$, with $t = \bar{d}/(s_d/\sqrt{n}) = 3/1 = 3.0$. On the 5 df row, 3.0 lies between 2.571 and 3.365, so the one-sided p-value is between 0.01 and 0.025 and the two-sided one is between 0.02 and 0.05 (software gives 0.030 two-sided). Both reject at 0.05.

Pairing matters here. Delays range from 28 to 65 ms across workloads, far more than the 3 ms effect. An unpaired Welch test on these same 12 numbers would leave that workload spread in the standard error and give $t \approx 0.43$ and $p \approx 0.68$ (computed by software, just to show the contrast), which misses the effect completely (Comp p.6-8). Note that workload 1 favors B, and that's fine, since the claim is about the mean difference. This is HW6 Problem 1 with six workloads.

B7. Unpaired comparison with Welch.

The transactions differ between systems and the counts differ, so no value of A goes with a value of B. The data are unpaired, and Welch's t is the default (Comp p.5, Comp p.12-13).

System A. The sum is 312, so $\bar{x}_A = 312/6 = 52$. Deviations are 0, 5, -6, 2, -4, 3, with squares 0, 25, 36, 4, 16, 9, summing to 90.

$$s_A^2 = \frac{90}{5} = 18, \qquad \frac{s_A^2}{n_A} = \frac{18}{6} = 3.$$

System B. The sum is 360, so $\bar{x}_B = 360/8 = 45$. Deviations are 0, -4, 2, -1, 5, -3, 1, 0, with squares 0, 16, 4, 1, 25, 9, 1, 0, summing to 56.

$$s_B^2 = \frac{56}{7} = 8, \qquad \frac{s_B^2}{n_B} = \frac{8}{8} = 1.$$

The standard error of the difference is

$$s = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{3 + 1} = 2.$$

The Welch degrees of freedom are (Comp p.12)

$$\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}} = \frac{(3 + 1)^2}{\frac{3^2}{5} + \frac{1^2}{7}} = \frac{16}{1.8 + 0.1429} = \frac{16}{1.9429} \approx 8.24.$$

Round down to 8, since the table has whole rows and rounding down is the safe direction. As a sanity check, ν must land between $\min(n_A, n_B) - 1 = 5$ and $n_A + n_B - 2 = 12$, and 8.24 does (Comp p.12).

The 90% interval for $\mu_A - \mu_B$ uses $t[0.05,\ 8] = 1.860$.

$$(\bar{x}_A - \bar{x}_B) \pm t[0.05,\ 8]\, s = 7 \pm 1.860(2) = 7 \pm 3.72 = (3.28,\ 10.72) \text{ ms}.$$

The interval excludes 0 and is positive, so at the 90% level A takes longer and B is faster, by somewhere between 3.3 and 10.7 ms of CPU time. As a test, $t = 7/2 = 3.5$, which is above $t[0.005,\ 8] = 3.355$, so the one-sided p-value is below 0.005.

The variance ratio is $18/8 = 2.25$, under 4, so the old rule of thumb would have pooled. The current advice is Welch by default, chosen before you see any p-value, because picking whichever version clears the cutoff is p-hacking (Comp p.13). This is HW6 Problem 2 with small samples.

B8. Measurement.

(a) One reading counts whole ticks, so it returns either 2 µs or 3 µs depending on where the operation starts relative to a tick, and never 2.6. The quantization error is up to one tick, 1 µs, which is about 38% of the thing you're measuring. That's the slides' $X \pm \Delta$ quantization error, and it's random (CI p.9-10).

(b) Subtract the loop's own cost, then divide by the count (measurement study guide, Part B).

$$\frac{27{,}412 - 1{,}412}{10{,}000} = \frac{26{,}000}{10{,}000} = 2.6 \text{ µs per operation.}$$

The one-tick error is now spread over 10,000 operations, so the resolution is $1/10{,}000 = 0.0001$ µs, or 0.1 ns. The assumption is that the operation behaves the same when repeated. It often doesn't, because the loop warms the cache and the branch predictor, and the compiler may even drop a loop body whose result is unused (measurement study guide, question 3).

(c) Two reads add $2c = 40$ ns, so the overhead fraction is $2c/(T + 2c)$.

$$T = 60 \text{ ns}: \ \frac{40}{60+40} = \frac{40}{100} = 40\%, \qquad T = 4000 \text{ ns}: \ \frac{40}{4000+40} = \frac{40}{4040} \approx 1\%.$$

The rule is that if an operation isn't one or two orders of magnitude slower than a clock read, you time a loop of them as in (b) instead (measurement study guide, question 4).

(d) The perturbation is $(86 - 80)/80 = 6/80 = 7.5\%$. The probes measured alone sum to 4.5 s, but together they cost 6 s, and that's not a contradiction, because perturbation isn't additive. Two probes' buffers can evict the program's working set and slow it down beyond their own cost, or the second probe can find data already cached and cost less. So you have to measure the configuration you actually ran, with and without instrumentation (Meas p.16-17).

(e) The variance of the overall mean with p processes of k iterations is $\sigma_p^2/p + \sigma_i^2/(pk)$ (measurement study guide, Part B).

$$1 \times 16: \ \frac{48}{1} + \frac{16}{16} = 49, \quad \text{SE} = 7 \text{ ms}$$

$$2 \times 8: \ \frac{48}{2} + \frac{16}{16} = 25, \quad \text{SE} = 5 \text{ ms}$$

$$16 \times 1: \ \frac{48}{16} + \frac{16}{16} = 4, \quad \text{SE} = 2 \text{ ms}$$

Same 16 measurements, but the standard error drops from 7 ms to 2 ms. Only more processes shrink the first term, so extra iterations inside one process can't buy down the bigger source of variance. Spend the budget at the level where the variance is (Meas p.25).

Part C

C1. The unit is one run, so n = 5 per system, and a "run" here is a separate measurement of one system, not a matched pair (HT p.20, Comp p.4). The headline comparison is unpaired, because run i of FastKV has nothing in common with run i of RocksDB, and they weren't even on the same day. The 13-workload comparison could have been paired by workload, since both systems can run the same workload, and pairing would cancel the large differences between workloads (Comp p.5-6).

C2. No. The bars are ±1 standard deviation, which describes the spread of single runs and doesn't shrink with n, so it says nothing direct about how precisely the means are known. The overlap rule on the slides is for confidence intervals, and even then non-overlap is the only safe direction (Comp p.15). The claim needs an interval on the difference, which C3 builds. The bars are 116.2 to 120.2 and 100.6 to 108.6, so they don't overlap, but that isn't the evidence.

C3. The claim is that FastKV has higher mean throughput, so $H_0 : \mu_F = \mu_R$ and $H_1 : \mu_F > \mu_R$, one-sided (HT p.4-5). For the interval, use Welch with $n = 5$ each (Comp p.12).

$$\frac{s_F^2}{n_F} = \frac{2.0^2}{5} = 0.8, \qquad \frac{s_R^2}{n_R} = \frac{4.0^2}{5} = 3.2, \qquad s = \sqrt{0.8 + 3.2} = \sqrt{4} = 2.$$

$$\nu = \frac{(0.8 + 3.2)^2}{\frac{0.8^2}{4} + \frac{3.2^2}{4}} = \frac{16}{0.16 + 2.56} = \frac{16}{2.72} \approx 5.88, \text{ round down to } 5.$$

$$13.6 \pm t[0.025,\ 5]\,(2) = 13.6 \pm 2.571(2) = 13.6 \pm 5.142 = (8.46,\ 18.74) \text{ Kops/s}.$$

The rewrite reads "FastKV's mean throughput on workload A was 13.6 Kops/s higher than RocksDB's (95% CI 8.5 to 18.7 Kops/s, Welch t, 5 runs each), which is roughly 8% to 18% of RocksDB's 104.6." The interval excludes 0, so the conclusion happens to survive,

but by correct reasoning instead of the bar picture (Comp p.24, Comp p.26). It's still only a statement about sampling error, and C6 is why that matters.

C4. No. Throughput is a rate, and the mean of rates for equal work should be the harmonic mean, because it equals total work over total time (Summ p.28, Summ p.32). An arithmetic mean of rates gives too much weight to the fast workloads. Better still, report the per-workload numbers so readers can pick their own summary.

C5. That's 13 tests at $\alpha = 0.05$ with no correction. Even if FastKV made no difference anywhere you'd expect $13 \times 0.05 = 0.65$ false positives, and the chance of at least one is $1 - 0.95^{13}$, roughly a coin flip (HT p.26-27). Four out of 13 is more than chance alone would give, but you can't say which four are real without a correction such as Bonferroni at $0.05/13 \approx 0.0038$ per test, or Holm (HT p.28). The paper should also report all 13 results, not just the ones that cleared 0.05 (Comp p.26).

C6. Any two of these. System and day are confounded, since every FastKV run was on Monday and every RocksDB run was on Tuesday after a kernel update. That's a systematic error that sits entirely outside the interval, and more runs would only tighten the interval around the biased difference. The fix is to interleave the two systems in randomized order on the same kernel (Meas p.20, Meas p.22-23). gettimeofday() isn't monotonic, so NTP can move it mid-run, and CLOCK_MONOTONIC is the right clock (Meas p.12). The 30-second warm-up cut is asserted, not justified, so you don't know how it was chosen or whether a steady state exists (Meas p.14).

C7. Two things. A large p-value isn't evidence of equivalence, especially with 5 runs, since the test may have had almost no power (HT p.14). And a t test compares means, while p99 is a quantile, so if the claim is about the tail the paper should compare p99s directly, for instance with a bootstrap interval on their difference (Comp p.18). An honest version would give the p99 difference with its interval and say how large a difference the experiment could have detected.

Practice set 2

A second mock exam on fresh numbers and on the topics set 1 left out, like weighted means, histograms, the variance interval, chi-square goodness of fit, two proportions, sample size and the sign test. Same rules as set 1. No electronics, pencil only, write each answer before you look at its solution. Aim for about 25 minutes on Part A and 70 minutes on Part B.

These are the only table values set 2 needs. The z and t brackets are upper-tail areas as before. The chi-square row gives the area *below* the value, which is how the slides label χ^2_L and χ^2_U (CI p.59-60).

Upper-tail area	**0.20**	**0.10**	**0.05**	**0.025**	**0.005**
z	0.842	1.282	1.645	1.960	2.576

Area below	**0.025**	**0.05**	**0.95**	**0.975**	**0.99**
χ^2, 3 df	0.22	0.35	7.81	9.35	11.34
χ^2, 9 df	2.70	3.33	16.92	19.02	21.67

Normal-table values $\Phi(2.0) = 0.9772$ and $\Phi(2.5) = 0.9938$. Powers of two $2^4 = 16$, $2^5 = 32$, $2^{10} = 1024$, $2^{16} = 65{,}536$ and $2^{32} = 4{,}294{,}967{,}296$.

Part A. Concepts (10 short questions, about 25 minutes)

Two or three sentences each, with the arithmetic where there is some.

A1. Three programs run at 100, 200 and 400 MFLOPS. (a) Each program runs for exactly 10 s. What's the mean MFLOPS? (b) Each program instead does the same 1000 million floating-point operations. What's the mean MFLOPS now? Say which mean you used each time and why they differ.

A2. True or false, with a reason. (a) A program that reports 0 MFLOPS does no useful work. (b) MFLOPS on machine A and machine B can be compared fairly because a FLOP is a FLOP.

A3. A paper compares old and new versions of a compiler on 5 benchmarks. The new version wins on all 5, and the paper reports "sign test, $p < 0.05$". Is that possible? What's the smallest two-sided p-value this design can produce?

A4. Five benchmarks give p-values 0.004, 0.011, 0.020, 0.035 and 0.30. At a family level of $\alpha = 0.05$, how many are significant with no correction, with Bonferroni, with Holm and with Benjamini-Hochberg?

A5. A free-running 32-bit counter ticks at 1 MHz. (a) How often does it roll over? (b) You read 4,294,966,296 at the start of an operation and 3,000 at the end. How long did it take, and what does a naive end minus start give? (c) With a 16-bit counter at the same rate, an operation really takes 100 ms. What elapsed time do you compute, and why can't you detect the mistake from the readings alone?

A6. (a) Why is the 95% confidence interval for σ^2 not centered on s^2? (b) The t interval for the mean gets rescued by the Central Limit Theorem when the data aren't normal. Does the variance interval get the same rescue?

A7. A power calculation says 20 runs per system will detect a 4 ms difference with power 0.80 at $\alpha = 0.05$. Your advisor wants to detect 2 ms instead. How many runs per system now? And after the experiment, a colleague offers to compute the power from your observed p-value. Is that useful?

A8. A sampling profiler takes 400 samples and only 3 land in `gc()`. Your friend reports the 95% interval

$$0.0075 \pm 1.96\sqrt{0.0075 \times 0.9925/400}.$$

What's wrong with relying on it, and what should be used instead?

A9. A CDN change cuts mean page load by 0.4 ms. The page-load standard deviation is 20 ms and the comparison used 10,000 loads. Compute Z and the two-sided p-value. Is the change significant, and is it important?

A10. Pick a test for each design and say why. (a) The same 12 inputs run on two versions, and the differences look symmetric but one is a big outlier. (b) The same 15 inputs on two versions, and the differences are strongly right-skewed. (c) 20 runs on machine A and 25 different runs on machine B, with long right tails. (d) 10 runs on each of two machines, roughly normal, but one machine's spread is three times the other's.

Part B. Problems (6 problems, about 70 minutes)

Show every step and name the formula before you plug in.

B1. A weighted harmonic mean. A simulation spends its floating-point work in three phases. Half of all FLOPs run in a dense phase at 400 MFLOPS, a quarter in a sparse phase at 200 MFLOPS and a quarter in a communication-bound phase at 100 MFLOPS. (a) What is the program's overall MFLOPS? Use the weighted mean that fits and say why. (b) What do the unweighted arithmetic mean and the FLOP-weighted arithmetic mean give, and why are they wrong? (c) Check your answer to (a) by assuming 1000 million FLOPs in total and adding up the time of each phase. (d) What fraction of the run's time does each phase take, and what does the time-weighted arithmetic mean of the rates give?

B2. Sturges and a histogram. Twenty response times in ms, in the order measured.

23	17	31	12	19	25	16	28	20	14
35	18	22	10	27	19	24	17	30	21

(a) How many bins does Sturges' rule give? (b) Using that many equal-width bins over the range, with bins of the form lo $<$ x $\leq$ hi as on the slides and the first bin also taking the minimum, give the count and relative frequency of each bin. (c) Describe the shape, and check your description against the mean and median. (d) A colleague redraws it with 10 bins of width 2.5. What changes, and what does that tell you? (e) How many bins would Sturges give for 1000 points?

B3. Chi-square goodness of fit. A load balancer is supposed to spread requests evenly over four servers. In a sample of 120 requests the servers received the following.

Server	**1**	**2**	**3**	**4**
Requests	42	20	34	24

(a) State H_0 and H_1, compute the expected counts and the χ^2 statistic, and give the degrees of freedom. (b) Test at $\alpha = 0.05$ and at $\alpha = 0.01$, and bracket the p-value. (c) If the claimed distribution had a parameter that you estimated from these same 120 requests, how would the degrees of freedom change? (d) Why does the textbook merge cells before computing χ^2 when some expected counts are small? (e) If the test had not rejected, could you report that the load is balanced?

B4. Confidence interval for a variance. Ten runs of a query take these times in ms.

44	46	47	47	50	50	52	53	55	56

(a) Find $\bar{x}$ and s^2. (b) Give the 95% confidence interval for σ^2 and for σ. (c) Give the 90% interval for σ^2. (d) The vendor's spec sheet says the run-to-run standard deviation is 3 ms. Do the data contradict it at the 95% level? (e) What assumption does the interval rest on, and how badly does it fail if the assumption is wrong?

B5. Comparing two proportions. Under the old cache configuration 180 of 800 requests missed a 100 ms latency target. Under the new configuration, on a separate set of requests, 140 of 800 missed it. (a) State H_0 and H_1 for "the miss rate changed" and run the two-proportion Z test. (b) Give the verdicts at $\alpha = 0.05$ and $\alpha = 0.01$ and the p-value. (c) Give a 95% confidence interval for the drop in miss rate. (d) Why does the test use a pooled proportion but the interval doesn't? (e) Check that the normal approximation is reasonable.

B6. Sample size and the sign test. You're planning to compare an old and a new build. (a) A pilot shows run-to-run standard deviation $s = 10$ ms on each build. With unpaired runs, how many runs per build do you need to detect a 5 ms difference at $\alpha = 0.05$ (two-sided) with power 0.80? (b) If you instead run both builds on the same inputs and the pilot shows the paired differences have $s_d = 4$ ms, how many pairs do you need? (c) Back to the unpaired design, how many runs per build to detect 2.5 ms? (d) You end up with 11 paired inputs, and the differences old minus new (ms) are listed below. Run a sign test at $\alpha = 0.05$, two-sided. (e) Why is the sign test the right choice here rather than the paired t test?

Input	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
d	3.1	0.4	12.0	0	1.2	−0.8	2.5	0.9	25.0	1.7	0.3

Practice set 2, solutions

Part A

A1. (a) Equal time means the rates get equal weight in total work. Total work is $10(100 + 200 + 400) = 7000$ million FLOPs in 30 s, so the rate is $7000/30 = 233.3$ MFLOPS, which is exactly the arithmetic mean $(100 + 200 + 400)/3$.

(b) Equal work is the setup the harmonic mean is built for, since it then equals total work over total time (Summ p.28, lecture 3 study guide).

$$\bar{x}_H = \frac{3}{\frac{1}{100} + \frac{1}{200} + \frac{1}{400}} = \frac{3}{\frac{4+2+1}{400}} = \frac{1200}{7} \approx 171.4 \text{ MFLOPS}.$$

Check it the long way. The programs take 10, 5 and 2.5 s, so 3000 million FLOPs take 17.5 s, and $3000/17.5 = 171.4$. The answers differ because with equal work the slow program runs longest and drags the rate down, while with equal time every rate counts for the same 10 s. A rate mean has to be inversely proportional to total time, and which mean does that depends on whether work or time was held fixed (Summ p.25).

A2. (a) False. An integer program reports 0 MFLOPS but still does useful work like sorting and searching (Metrics p.19).

(b) False. There's too much freedom in how to count a FLOP, for example transcendental operations and square roots, so MFLOPS isn't consistent across machines (Metrics p.19).

A3. It isn't possible. Under H_0 the number of wins S is Binomial(5, $\frac{1}{2}$), so all 5 wins has probability 1/32, and the two-sided p-value is $2/32 = 0.0625$. That's the smallest p any outcome can give, and it's above 0.05. With 5 or fewer pairs the sign test can't reach significance at all (Comp p.22-23, comparing-alternatives study guide).

A4. No correction rejects every p below 0.05, which is 4 of them. Bonferroni tests each at $0.05/5 = 0.01$, so only 0.004 passes, which is 1. Holm compares the sorted p-values with $\alpha/(m - k + 1)$. First $0.004 \le 0.05/5 = 0.01$, reject. Then $0.011 \le 0.05/4 = 0.0125$, reject. Then $0.020 > 0.05/3 = 0.0167$, so stop, which gives 2. Benjamini-Hochberg compares $p_{(k)}$ with $k\alpha/m$, which is 0.01, 0.02, 0.03, 0.04, 0.05. The largest k that passes is $k = 4$, since $0.035 \le 0.04$, so it rejects 4. Same p-values, and the count runs from 1 to 4, exactly the point of the slides' six-comparison table (HT p.28-29). BH controls the false discovery rate, not the chance of any false positive, so it isn't the same guarantee as the other two.

A5. (a) It rolls over every 2^{32} ticks, which at 1 MHz is 4,294,967,296 µs, about 4295 s or 71.6 minutes.

(b) The counter wrapped. From the start reading to the wrap is $4{,}294{,}967{,}296 - 4{,}294{,}966{,}296 = 1{,}000$ ticks, then 3,000 more, so the operation took 4,000 µs, which is 4 ms. A naive $3{,}000 - 4{,}294{,}966{,}296$ is a huge negative number. Doing the subtraction in unsigned 32-bit arithmetic, which is modulo 2^{32}, gives the right 4,000, as long as the interval is shorter than one period.

(c) The 16-bit counter wraps every $2^{16} = 65{,}536$ µs, about 65.5 ms. A 100 ms operation wraps once, and the modulo subtraction gives $100 - 65.536 = 34.464$ ms. Both readings look perfectly normal, so nothing in the data shows the lost period. The timer's period has to be longer than anything you time, or you have to count the overflows. It's one more reason to report which clock you used, the way the slides ask you to report its resolution (Meas p.12).

A6. (a) The interval divides $(n - 1)s^2$ by two chi-square critical values, and the chi-square distribution isn't symmetric (CI p.59-60). The lower critical value sits much closer to 0 than the upper one sits above the middle, so the upper end of the interval lands much farther from s^2 than the lower end does. B4 shows it, with $s^2 = 16$ and an interval of 7.6 to 53.3.

(b) No. The Central Limit Theorem is about sums and means of many values (CI p.64), so it rescues the t interval. The variance interval leans on the data themselves being normal, and the slides warn that for non-normal populations it isn't very accurate, especially for small samples (CI p.62).

A7. The required n grows with $(\sigma/\delta)^2$, so halving δ costs four times the runs, $4 \times 20 = 80$ per system (HT p.19). The post-hoc power isn't useful. Power computed after the fact from your own p-value tells you nothing new, and the calculation belongs before the data are collected (HT p.19, Comp p.25).

A8. The interval is the Wald interval, and it needs np and $n(1 - p)$ both comfortably large, but here $np = 3$ (CI p.47, confidence-intervals study guide). It even breaks visibly. $\sqrt{0.0075 \times 0.9925/400} \approx 0.00431$ and $1.96 \times 0.00431 \approx 0.0085$, so the lower end is $0.0075 - 0.0085 \approx -0.001$, a negative fraction of time. Use the Wilson or Agresti-Coull interval instead. The slides recommend them over Wald, Wilson for $n < 40$ and Agresti-Coull above (CI p.49). Or collect many more samples so the count in `gc()` isn't tiny.

A9.

$$Z = \frac{\text{effect}}{\sigma/\sqrt{n}} = \frac{0.4}{20/\sqrt{10{,}000}} = \frac{0.4}{0.2} = 2.0, \qquad p = 2\big(1 - \Phi(2.0)\big) = 2(1 - 0.9772) = 0.0456.$$

It's significant at 0.05, but the effect is 0.4 ms against a 20 ms spread, a standardized size of $0.4/20 = 0.02$. No user would notice that. With enough data every non-zero effect becomes significant, so significance is partly a statement about n (HT p.17). Report the 0.4 ms with its interval and let the reader judge whether it matters (HT p.18).

A10. (a) Wilcoxon signed-rank. The design is paired, and the rank test is robust to an outlier as long as the differences are roughly symmetric (Comp p.17, Comp p.21).

(b) The sign test or a bootstrap, or analyze log(A/B). Signed-rank assumes symmetric differences, so skew rules it out (Comp p.17-18, Comp p.21, Comp p.23).

(c) Mann-Whitney U or a bootstrap. The samples are unpaired and skewed (Comp p.17, Comp p.19).

(d) Welch's t. It's unpaired and roughly normal, and Welch doesn't care about unequal variances, which is why it's the default (Comp p.12-13, Comp p.17).

Part B

B1. A weighted harmonic mean.

(a) These are rates, and the weights are shares of the work, not of the time. The mean rate has to be inversely proportional to total

time, so use the weighted harmonic mean $\bar{x}_H = 1/\sum(w_i/x_i)$ with $\sum w_i = 1$ (Summ p.25, Summ p.28, Summ p.33).

$$\bar{x}_H = \frac{1}{\frac{0.5}{400} + \frac{0.25}{200} + \frac{0.25}{100}} = \frac{1}{0.00125 + 0.00125 + 0.0025} = \frac{1}{0.005} = 200 \text{ MFLOPS}.$$

(b) The unweighted arithmetic mean is $(400 + 200 + 100)/3 = 233.3$ MFLOPS. The FLOP-weighted arithmetic mean is $0.5(400) + 0.25(200) + 0.25(100) = 200 + 50 + 25 = 275$ MFLOPS. Both overstate the real rate because they ignore that the slow phase eats the most time. An arithmetic mean of rates isn't proportional to anything physical unless the weights are shares of time, which (d) shows.

(c) Take 1000 million FLOPs. The dense phase does 500 at 400 MFLOPS, so $500/400 = 1.25$ s. The sparse phase does 250 at 200, so 1.25 s. The communication phase does 250 at 100, so 2.5 s. The total is 5 s, and $1000/5 = 200$ MFLOPS, matching (a). That's the lecture 3 study guide's point that the work-weighted harmonic mean is just total work over total time written as a mean.

(d) The phases take $1.25/5 = 0.25$, $1.25/5 = 0.25$ and $2.5/5 = 0.5$ of the time.

$$0.25(400) + 0.25(200) + 0.5(100) = 100 + 50 + 50 = 200 \text{ MFLOPS}.$$

So the arithmetic mean weighted by time and the harmonic mean weighted by work give the same answer, because both equal total work over total time. The mistake in (b) was weighting an arithmetic mean by work.

B2. Sturges and a histogram.

(a) Sturges' rule is $k = 1 + \log_2 n$ (Summ p.54). Since $16 < 20 < 32$, $\log_2 20$ is between 4 and 5, about 4.32, so $k \approx 5.32$. The slide floors it and HW3 rounded it, and both give 5 bins here.

(b) The range is $35 - 10 = 25$, so 5 bins are $25/5 = 5$ ms wide. Sorted, the data are 10, 12, 14, 16, 17, 17, 18, 19, 19, 20, 21, 22, 23, 24, 25, 27, 28, 30, 31, 35.

Bin (ms)	$10 \le x \le 15$	$15 < x \le 20$	$20 < x \le 25$	$25 < x \le 30$	$30 < x \le 35$
Values	10, 12, 14	16, 17, 17, 18, 19, 19, 20	21, 22, 23, 24, 25	27, 28, 30	31, 35
Count	3	7	5	3	2
Relative frequency	15%	35%	25%	15%	10%

The counts add to 20 and the percentages to 100%. The relative frequency is the count divided by $n = 20$, as the slides plot it (Summ p.54-55).

(c) It peaks in the 15 to 20 bin and tails off further to the right than to the left, so it's mildly right-skewed. The sum is 428, so the mean is $428/20 = 21.4$ ms. The median is the average of the 10th and 11th values, $(20 + 21)/2 = 20.5$ ms (Summ p.14), and the course convention gives the same, since $r = 20(0.5) + 0.5 = 10.5$. The mean sits above the median, which is what a right tail does.

(d) With 2.5 ms bins the counts become 2, 1, 3, 4, 2, 3, 1, 2, 1, 1. The single peak breaks into bumps that are just noise from having about two points per bin. Same data, different cell size, different shape, which is why the slides say cell size is critical (Summ p.54, Summ p.56).

(e) $2^{10} = 1024$, so $\log_2 1000$ is just under 10, about 9.97, and $k \approx 10.97$. The slide's floor gives 10 bins and rounding gives 11. Say which convention you used.

B3. Chi-square goodness of fit.

(a) H_0 says each server gets a quarter of the requests, and H_1 says the shares aren't all equal. The expected count is $120/4 = 30$ for every server, which is well above 5. The statistic is $\chi^2 = \sum(o_i - e_i)^2/e_i$ (chi-square goodness-of-fit handout).

$$\chi^2 = \frac{(42-30)^2 + (20-30)^2 + (34-30)^2 + (24-30)^2}{30} = \frac{144 + 100 + 16 + 36}{30} = \frac{296}{30} \approx 9.87.$$

The degrees of freedom are $k - m$, where $k = 4$ cells and m is the number of quantities taken from the data to compute the expected counts. Only the total of 120 was used, so $m = 1$ and df $= 3$.

(b) The test rejects for large χ^2 only. At $\alpha = 0.05$ the cutoff is the 0.95 value on 3 df, 7.81, and $9.87 > 7.81$, so reject. At $\alpha = 0.01$ the cutoff is 11.34, and $9.87 < 11.34$, so don't reject. The p-value is between 0.01 and 0.05 (software gives 0.020).

(c) Each parameter estimated from the same data adds one to m, so df would drop to $4 - 2 = 2$ (handout).

(d) The χ^2 statistic only follows the chi-square distribution approximately, and the approximation is poor when expected counts are small. The handout's Poisson example merges the end cells so that no expected frequency is below 5.

(e) No. Failing to reject isn't evidence for H_0, only an absence of evidence against it (HT p.4, HT p.14). You'd report the observed shares and how big an imbalance 120 requests could have detected.

B4. Confidence interval for a variance.

(a) The sum is 500, so $\bar{x} = 50$ ms. The deviations are $-6, -4, -3, -3, 0, 0, 2, 3, 5, 6$, with squares 36, 16, 9, 9, 0, 0, 4, 9, 25, 36, summing to 144.

$$s^2 = \frac{144}{10 - 1} = 16 \text{ ms}^2, \qquad s = 4 \text{ ms}.$$

(b) The interval is $(n-1)s^2/\chi^2_U \le \sigma^2 \le (n-1)s^2/\chi^2_L$ on $n - 1 = 9$ df (CI p.59). Here $(n-1)s^2 = 144$, and for 95% the critical values are the 0.025 and 0.975 points, $\chi^2_L = 2.70$ and $\chi^2_U = 19.02$.

$$\frac{144}{19.02} \approx 7.57 \le \sigma^2 \le \frac{144}{2.70} \approx 53.33 \text{ ms}^2.$$

Taking square roots, $2.75 \le \sigma \le 7.30$ ms. Note the upper end is divided by the small critical value, and it's lopsided, from 8.4 below s^2 to 37.3 above it (CI p.60).

(c) For 90% use the 0.05 and 0.95 points, 3.33 and 16.92.

$$\frac{144}{16.92} \approx 8.51 \le \sigma^2 \le \frac{144}{3.33} \approx 43.24 \text{ ms}^2.$$

Software with unrounded critical values gives 43.31 for the upper end, so the gap is just table rounding.

(d) No. A standard deviation of 3 ms means $\sigma^2 = 9$, and 9 lies inside (7.57, 53.33). Equivalently 3 is inside (2.75, 7.30). Ten runs pin σ down only to somewhere between about 2.8 and 7.3 ms, so they can't rule out the spec.

(e) It assumes the run times themselves are normal (CI p.59). If they aren't, the interval can be badly off, especially at small n (CI p.62), and unlike the t interval for the mean the Central Limit Theorem doesn't fix it. Heavy-tailed run times, which are common, make it too narrow.

B5. Comparing two proportions.

(a) $\hat{p}_{\text{old}} = 180/800 = 0.225$ and $\hat{p}_{\text{new}} = 140/800 = 0.175$. H_0 says $p_{\text{old}} = p_{\text{new}}$ and H_1 says $p_{\text{old}} \neq p_{\text{new}}$, two-sided because the question is whether the rate changed. Under H_0 both groups share one rate, so pool them, $\hat{p} = (180 + 140)/1600 = 0.2$. The statistic is $Z = (\hat{p}_1 - \hat{p}_2)/\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$ (hypothesis-testing study guide).

$$\text{SE} = \sqrt{0.2 \times 0.8 \left(\frac{1}{800} + \frac{1}{800}\right)} = \sqrt{0.16 \times 0.0025} = \sqrt{0.0004} = 0.02, \qquad Z = \frac{0.225 - 0.175}{0.02} = \frac{0.05}{0.02} = 2.5.$$

(b) $|Z| = 2.5 > 1.960$, so reject at 0.05. $2.5 < 2.576$, so don't reject at 0.01. The two-sided p-value is $2(1 - 0.9938) = 0.0124$.

(c) Each rate's variance is $p(1-p)/n$ as on CI p.47, and the variances of two independent estimates add, the same way they do in Welch's standard error (Comp p.12).

$$\text{SE} = \sqrt{\frac{0.225 \times 0.775}{800} + \frac{0.175 \times 0.825}{800}} = \sqrt{\frac{0.174375 + 0.144375}{800}} = \sqrt{\frac{0.31875}{800}} \approx \sqrt{0.000398} \approx 0.0200.$$

$$0.05 \pm 1.96(0.0200) = 0.05 \pm 0.039 = (0.011,\ 0.089).$$

The new configuration cuts the miss rate by somewhere between 1.1 and 8.9 percentage points. The point estimate, 5 points out of 22.5, is a relative drop of about 22%.

(d) The test computes the standard error assuming H_0 is true, and under H_0 there's one common rate, so the best estimate of it uses all 1600 requests. The interval estimates the difference without assuming it's zero, so each group keeps its own rate.

(e) The counts are 180 and 620 for the old configuration and 140 and 660 for the new one, all far above 10, so the normal approximation is fine (CI p.47, confidence-intervals study guide). The two groups are separate sets of requests, so this is the independent two-proportion test. If both configurations had been judged on the same requests you'd need McNemar's test instead (hypothesis-testing study guide).

B6. Sample size and the sign test.

(a) For unpaired runs, n per group $= 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$ (Comp p.25). Power 0.80 means $\beta = 0.20$ and $z[0.20] = 0.842$, which the slides round to 0.84 (HT p.19).

$$n = 2\left(\frac{(1.96 + 0.84) \times 10}{5}\right)^2 = 2(5.6)^2 = 2(31.36) = 62.72.$$

Round up to 63 runs per build, 126 in all.

(b) For pairs, $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$ (Comp p.25).

$$n = \left(\frac{2.8 \times 4}{5}\right)^2 = (2.24)^2 = 5.02.$$

Round up to 6 pairs. Pairing removes the input-to-input variation, so the spread that matters drops from 10 to 4 and the factor of 2 disappears. That's 6 pairs against 126 runs.

(c) Halving δ multiplies n by four (HT p.19).

$$n = 2\left(\frac{2.8 \times 10}{2.5}\right)^2 = 2(11.2)^2 = 2(125.44) = 250.88,$$

so 251 runs per build.

(d) Drop the zero difference on input 4, which leaves $n = 10$. Every remaining difference is positive except input 6, so $S = 9$. Under H_0 the median difference is 0 and S is Binomial(10, $\frac{1}{2}$) (Comp p.23).

$$P(S \ge 9) = \frac{\binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{10 + 1}{1024} = \frac{11}{1024} \approx 0.0107, \qquad p = 2 \times \frac{11}{1024} = \frac{22}{1024} \approx 0.0215.$$

$0.0215 < 0.05$, so reject. The new build is faster on more inputs than chance would explain.

(e) The differences are strongly right-skewed. Two of them, 12.0 and 25.0, are far above the rest, which sit between -0.8 and 3.1. The mean difference is $46.3/11 \approx 4.2$ ms while the median is 1.2 ms. Paired t assumes roughly normal differences, and signed-rank assumes symmetric ones, so for skewed paired differences the slides send you to the sign test or a bootstrap (Comp p.17, Comp p.21). The two big values also inflate s_d, and a paired t on these 11 differences gives only $t \approx 1.81$, $p \approx 0.10$ (software, just to show the contrast). That pick has to come from the design and a plot before any p-value is computed, since choosing whichever test clears 0.05 is p-hacking (Comp p.18). The sign test's claim is also narrower. It says the new build wins on most inputs, not how much faster it is on average.